



TECNOLOGICO NACIONAL DE MEXICO

INSTITUTO TECNOLÓGICO de Tuxtepec

Presenta

Alumno: David Sandoval Aguirre

Carrera

ING. SISTEMAS COMPUTACIONALES

Materia:

CIENCIAS DE DATOS

Docente:

JULIO AGUILAR CARMONA

Semestre: 8° A

Tuxtepec, Oax.

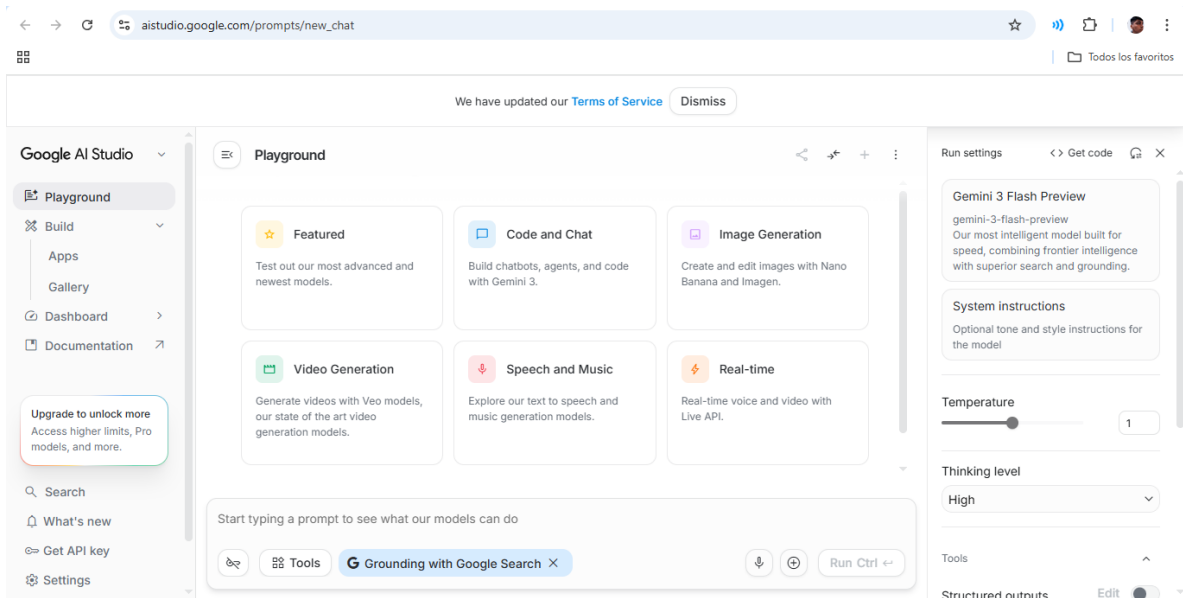
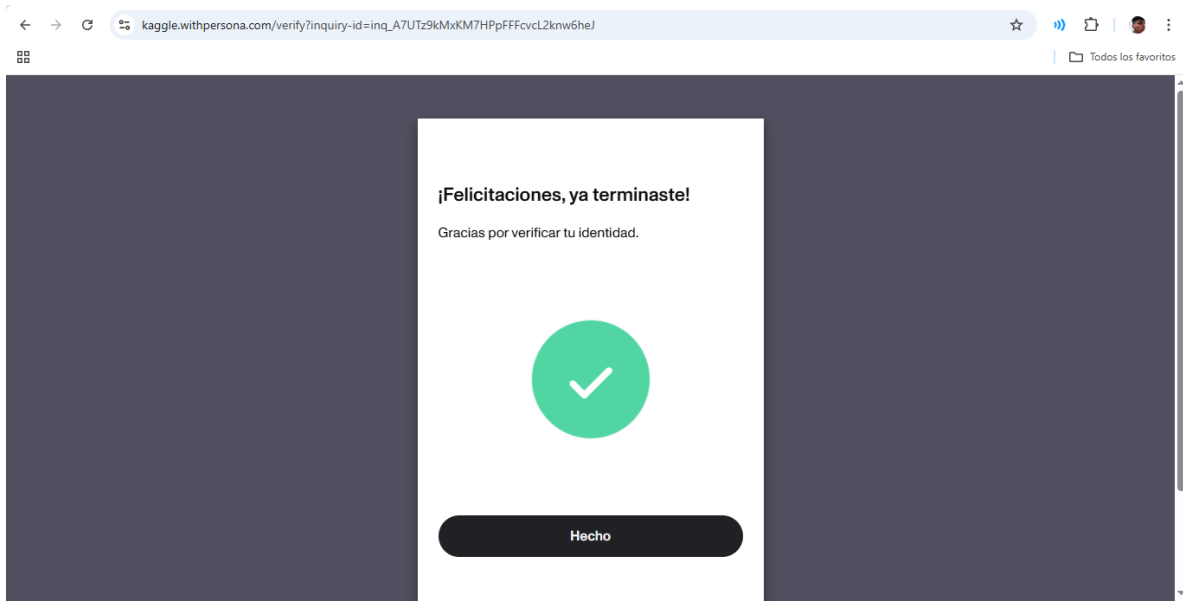
05/05/2026



Introducción

The image shows two screenshots. The top screenshot is of the Kaggle website's user dashboard for 'SANDOVAL Aguirre David'. The dashboard includes a sidebar with navigation links like 'Crear', 'Hogar', 'Competiciones', and 'Puntos de referencia'. The main content area displays a welcome message and statistics for various Kaggle features: 'conjuntos de datos' (0 created), 'Cuadernos' (0 created), 'Competiciones' (0 joined), 'Discusiones' (0 published), and 'Cursos' (0 completed). A progress indicator shows '0% a Experto'.

The bottom screenshot is of a YouTube video player showing a video titled 'How to use Kaggle Notebooks'. The video player includes a progress bar at 0:02 / 4:18 and a channel name 'Kaggle' with 323 k subscribers. To the right of the video player, there is a sidebar with recommended videos, including one from 'designrr' about creating ebooks and another about quantum computing.



← → ↻ aistudio.google.com/app/api-keys?project=gen-lang-client-0532772298

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Google AI Studio

< Panel

Claves API

Proyectos

Uso

Límite de frecuencia

Gasto

Facturación

Registros y conjuntos de datos

Registro de cambios ↗

🔍 Buscar

🔔 Qué hay de nuevo

➞ Obtén la clave API

⚙ Ajustes

Claves de API

Agrupar por **Clave API**

Crear clave de API

Filtrar por Todos los proyectos

Detalles de la clave de API

Clave de API
AlzaSyCWj8yYudMhLR1gNHL4S5V7LnC9BwZNaj0

Nombre
Curso de IA de 5 días

Nombre del proyecto
proyectos/967436590123

Número del proyecto
967436590123

[Borrar clave](#) [Copiar guía de inicio rápido de cURL](#) [Copiar clave](#)

¿No encuentras tus claves de API aquí?

Esta lista solo muestra las claves de API de los proyectos importados a Google AI Studio. Importa otros proyectos para administrar sus claves de API asociadas. También puedes crear una nueva clave de API arriba. [Más información](#)

← → ↻ aistudio.google.com/app/api-keys?project=gen-lang-client-0532772298

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Google AI Studio

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Claves API

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🔔 Qué hay de nuevo

➞ Obtén la clave API

⚙ Ajustes

Claves de API

Guía de inicio rápido de la API

Crear clave de API

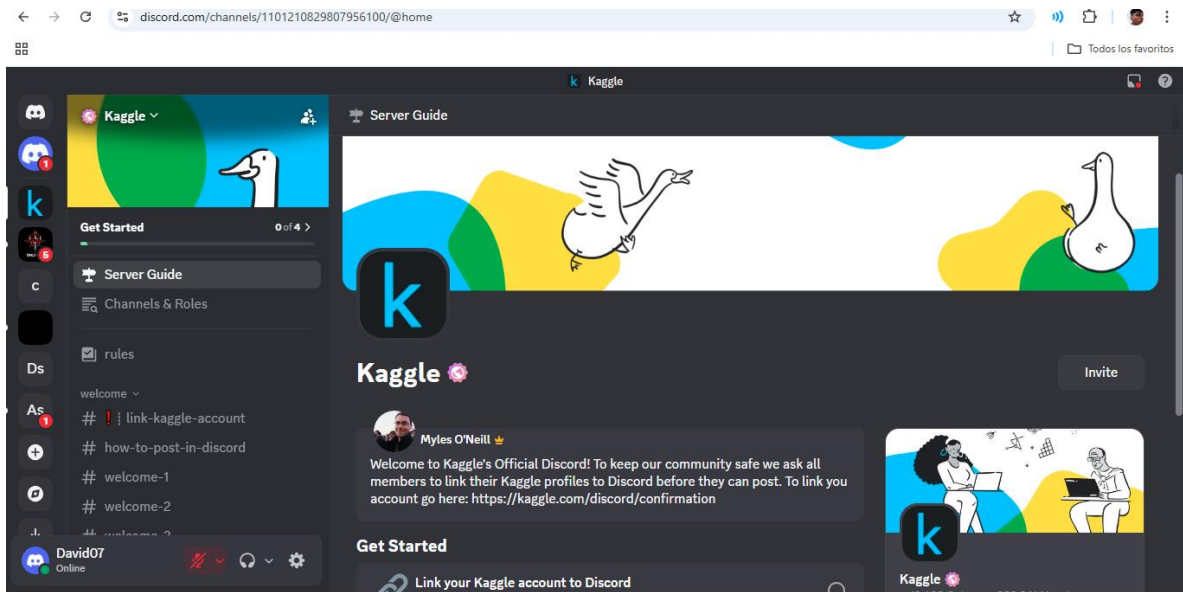
Agrupar por **Clave API** Proyecto

Filtrar por Todos los proyectos

Clave	Proyecto	Creado	Nivel de facturación
...Naj0 Curso de IA de 5 días	Proyecto Gemini predeterminado gen-lang-client-0532772298	30 de abril de 2026	Configurar la facturación Nivel gratuito
...Fct8 Clave API de Gemini predeterminada	Proyecto Gemini predeterminado gen-lang-client-0532772298	30 de abril de 2026	Configurar la facturación Nivel gratuito

¿No encuentras tus claves de API aquí?

Esta lista solo muestra las claves de API de los proyectos importados a Google AI Studio. Importa otros proyectos para administrar sus claves de API asociadas. También puedes crear una nueva clave de API arriba. [Más información](#)



Dia 1 A

youtube.com/watch?v=zTxvGzpfF-g

YouTube MX

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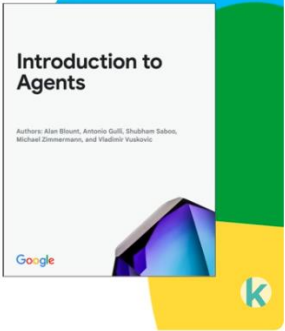
inglés español

Google Translate

Todos los favoritos

kaggle 5-Day AI Agents Intensive Course

Whitepaper Companion Podcast



Podcast complementario del informe técnico: Introducción a los agentes

Kaggle 323 mil suscriptores

Suscribirse

21 k

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Preguntar

Podcast complementario del informe técnico: Herramientas...

Podcast complementario del informe técnico: del prototipo...

Cómo creamos agentes eficaces: Barry Zhang, ...

Podcast complementario del informe técnico: Calidad del ...

kaggle.com/whitepaper-introduction-to-agents

Buscar

Introduction to Agents

Authors: Alan Blount, Antonio Gulli, Shubham Saboo, Michael Zimmermann, and Vladimir Vuskovic

Crear

Hogar

Competiciones

Puntos de referencia

Arena de juegos

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Más

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kaggle.com/code/sandovalaguirredavid/day-1a-from-prompt-to-action/edit

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Todos los favoritos

Day 1a - From Prompt to Action Borrador guardado

Archivo Editar Vista Correr Ajustes Complementos Ayuda

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+ Code + Markdown

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WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
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limitations under the License.

+ Code + Markdown

🚀 Your First AI Agent: From Prompt to Action

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Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr

Copiar al portapapeles

☒ GOOGLE_API_KEY *****

+ Agregar secreto Copiar fragmento

← → ↻

kaggle.com/code/sandovalaguirredavid/day-1a-from-prompt-to-action/edit

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Todos los favoritos

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Archivo Editar Vista Correr Ajustes Complementos Ayuda

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3. Authenticate in the notebook

Run the cell below to complete authentication.

+ Code + Markdown

▶

import os
from kaggle_secrets import UserSecretsClient

try:
 GOOGLE_API_KEY = UserSecretsClient().get_secret("GOOGLE_API_KEY")
 os.environ["GOOGLE_API_KEY"] = GOOGLE_API_KEY
 print("✅ Gemini API key setup complete.")
except Exception as e:
 print(
 f"🚨 Authentication Error: Please make sure you have added 'GOOGLE_API_KEY' to yo
)

✅ Gemini API key setup complete.

+ Code + Markdown

🚀 Your First AI Agent: From Prompt to Action

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Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr

Copiar al portapapeles

☒ GOOGLE_API_KEY *****

+ Agregar secreto Copiar fragmento

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Run All

Code

Sesión de borrador Cargando0 m

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```
from google.adk.agents import Agent
from google.adk.models.google_llm import Gemini
from google.adk.runners import InMemoryRunner
from google.adk.tools import google_search
from google.genai import types

print("✅ ADK components imported successfully.")
```

✅ ADK components imported successfully.

+ Code + Markdown

1.4: Helper functions

We'll define some helper functions. If you are running this outside the Kaggle environment, you don't need to do this.

```
[ ]: # Define helper functions that will be reused throughout the notebook
```

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

☑ GOOGLE_API_KEY *****

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Run All

Code

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```
</ol>
<em style="font-size: 0.9em; color: #555;">(If you click the button before run
</div>
<a href='(url)' target='_blank' style="
display: inline-block; background-color: #1a73e8; color: white; padding: 10px
text-decoration: none; border-radius: 25px; font-family: sans-serif; font-we
box-shadow: 0 2px 5px rgba(0,0,0,0.2); transition: all 0.2s ease;">
Open ADK Web UI (after running cell below) </
</a>
</div>
</div>

display(HTML(styled_html))

return url_prefix

print("✅ Helper functions defined.")
```

✅ Helper functions defined.

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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☑ GOOGLE_API_KEY *****

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Escribe aquí para buscar.

Kaggle Notebook E... Documento1 - Word

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Sesión de borrador Cargando0 m)

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When working with LLMs, you may encounter transient errors like rate limits or temporary service unavailability. Retry options automatically handle these failures by retrying the request with exponential backoff.

▶

```
retry_config=types.HttpRetryOptions(
    attempts=5, # Maximum retry attempts
    exp_base=7, # Delay multiplier
    initial_delay=1, # Initial delay before first retry (in seconds)
    http_status_codes=[429, 500, 503, 504] # Retry on these HTTP errors
)
```

+ Code + Markdown

🤖 Section 2: Your first AI Agent with ADK

😞 2.1 What is an AI Agent?

You've probably used an LLM like Gemini before, where you give it a prompt and it gives you a text response.

🔍

🗂

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr
```

Copiar al portapapeles

☒ GOOGLE_API_KEY

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Sesión de borrador Cargando0 m)

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```
root_agent = Agent(
    name="helpful_assistant",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    description="A simple agent that can answer general questions.",
    instruction="You are a helpful assistant. Use Google Search for current info or if un",
    tools=[google_search],
)

print("✅ Root Agent defined.")
```

+ Code + Markdown

✅ Root Agent defined.

2.3 Run your agent

New! It's time to bring your agent to life and send it queries. To do this, you need a "Runner", which is the central

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr
```

Copiar al portapapeles

☒ GOOGLE_API_KEY

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a. Create an `InMemoryRunner` and tell it to use our `root_agent`:

runner = InMemoryRunner(agent=root_agent)

print(✅ Runner created.)

✅ Runner created.

+ Code + Markdown

👉 Note that we are using the Python Runner directly in this notebook. You can also run agents using ADK command-line tools such as `adk run`, `adk web`, or `adk api_server`. To learn more, check out the documentation related to runtime in ADK.

+ Code + Markdown

b. Now you can call the `.run_debug()` method to send our prompt and get an answer.

👉 This method abstracts the process of session creation and maintenance and is used in prototyping. We'll explore "what sessions are and how to create them" on Day 3.

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0

Copiar al portapapeles

GOOGLE_API_KEY

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response = await runner.run_debug(
"What is Agent Development Kit from Google? What languages is the SDK available in?"
)

Created new session: debug_session_id

User > What is Agent Development Kit from Google? What languages is the SDK available in?
helpful_assistant > The Google Agent Development Kit (ADK) is an open-source framework designed to simplify the development and deployment of AI agents and multi-agent systems. It provides a foundation for building production-ready agentic applications with greater flexibility and control. Instead of treating large language models as simple request-response systems, ADK introduces an event-driven runtime architecture that orchestrates agents, tools, and persistent state into cohesive applications. This event-loop architecture allows applications to react to intermediate steps, such as tool invocations, and maintain conversational context across multiple exchanges. ADK is optimized for Google Cloud and Gemini models but is model-agnostic and can be deployed in various environments, including Google Cloud Run or Kubernetes.

The ADK SDK is available in the following languages:
* Python
* Go
* Java
* TypeScript

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0

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Documento1 - Word

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```
response = await runner.run_debug("What's the weather in London?")
```

Continue session: debug_session_id

User > What's the weather in London?
helpful_assistant > The weather in London is currently sunny with a temperature of 70°F (21°C). The humidity is around 31%. It feels like 74°F (23°C).

Here's a brief forecast for the coming days:
* **Thursday, April 30:** Sunny during the day and clear at night, with temperatures between 51°F (11°C) and 71°F (22°C).
* **Friday, May 1:** Sunny during the day and cloudy at night, with a 15-20% chance of rain and temperatures between 51°F (11°C) and 75°F (24°C).
* **Saturday, May 2:** Light rain is expected with a 35-70% chance of precipitation and temperatures between 52°F (11°C) and 66°F (19°C).

+ Code + Markdown

Section 3: Try the ADK Web Interface

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecretsClient
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

GOOGLE_API_KEY *****

+ Agregar secreto Copiar fragmento

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To use the ADK web UI, you'll need to create an agent with Python files using the `adk create` command.

Run the command below to generate a `sample-agent` folder that contains all the necessary files, including `agent.py` for your code, an `.env` file with your API key pre-configured, and an `__init__.py` file:

```
!adk create sample-agent --model gemini-2.5-flash-lite --api_key $GOOGLE_API_KEY
```

Agent created in /kaggle/working/sample-agent:
- .env
- __init__.py
- agent.py

+ Code + Markdown

Get your custom URL to access the ADK web UI in the Kaggle Notebooks environment:

```
[ ]: url_prefix = get_adk_proxy_url()
```

Misterios

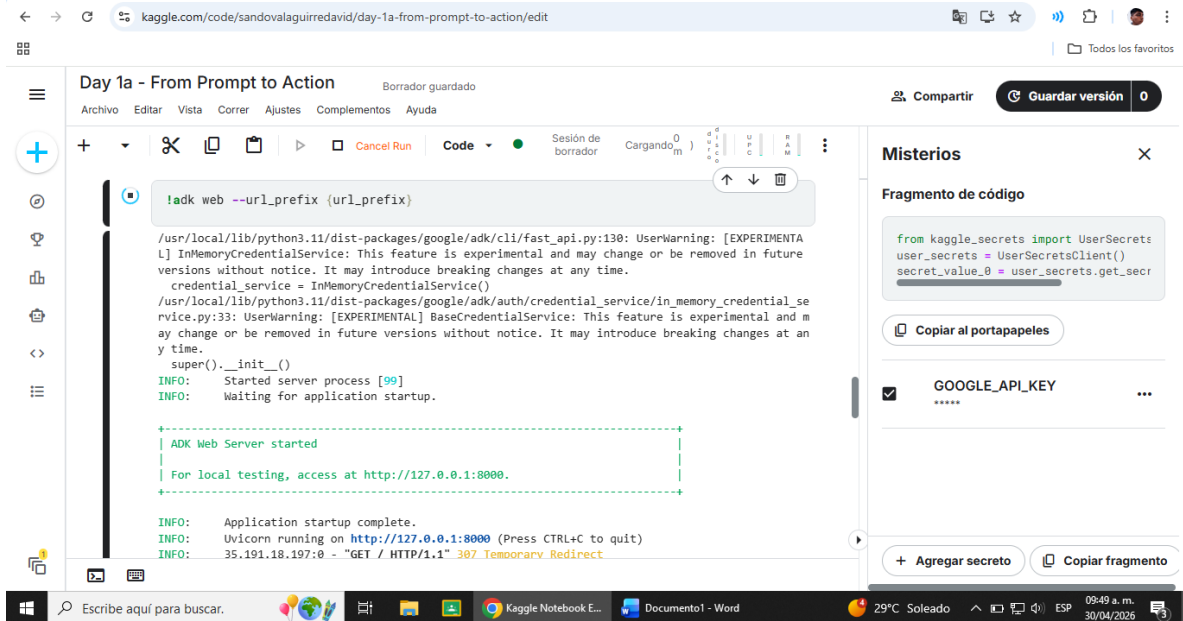
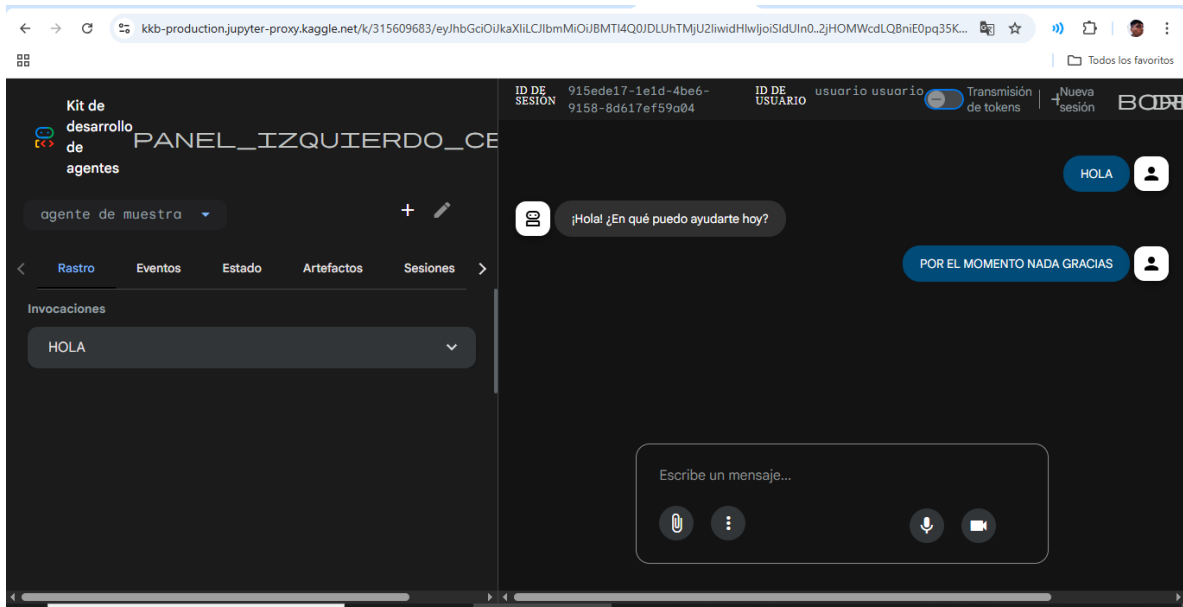
Fragmento de código

```
from kaggle_secrets import UserSecretsClient
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

GOOGLE_API_KEY *****

+ Agregar secreto Copiar fragmento



Dia 1 B

The image displays two sequential screenshots of a Kaggle notebook titled "Day 1b - Agent Architectures".

Top Screenshot:

- Section:** 1.3: Import ADK components
- Text:** "Now, import the specific components you'll need from the Agent Development Kit and the Generative AI library. This keeps your code organized and ensures we have access to the necessary building blocks."
- Code Cell:**

```
import os
from kaggle_secrets import UserSecretsClient

try:
    GOOGLE_API_KEY = UserSecretsClient().get_secret("GOOGLE_API_KEY")
    os.environ["GOOGLE_API_KEY"] = GOOGLE_API_KEY
    print("✅ Gemini API key setup complete.")
except Exception as e:
    print(f"🚨 Authentication Error: Please make sure you have added 'GOOGLE_API_KEY' to yo")
```

Output:
✅ Gemini API key setup complete.

Bottom Screenshot:

- Section:** 1.3: Import ADK components
- Text:** "Now, import the specific components you'll need from the Agent Development Kit and the Generative AI library. This keeps your code organized and ensures we have access to the necessary building blocks."
- Code Cell:**

```
from google.adk.agents import Agent, SequentialAgent, ParallelAgent, LoopAgent
from google.adk.models.google_llm import Gemini
from google.adk.runners import InMemoryRunner
from google.adk.tools import AgentTool, FunctionTool, google_search
from google.genai import types

print("✅ ADK components imported successfully.")
```

Output:
✅ ADK components imported successfully.

← → ↻ kaggle.com/code/sandovalaguirredavid/day-1b-agent-architectures/edit

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Day 1b - Agent Architectures

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1.4: Configure Retry Options

When working with LLMs, you may encounter transient errors like rate limits or temporary service unavailability. Retry options automatically handle these failures by retrying the request with exponential backoff.

```
retry_config=types.HttpRetryOptions(
    attempts=5, # Maximum retry attempts
    exp_base=7, # Delay multiplier
    initial_delay=1,
    http_status_codes=[429, 500, 503, 504], # Retry on these HTTP errors
)
```

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Section 2: Why Multi-Agent Systems? + Your First Multi-Agent

Compartir Guardar versión 0

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

GOOGLE_API_KEY

+ Agregar secreto Copiar fragmento

← → ↻ kaggle.com/code/sandovalaguirredavid/day-1b-agent-architectures/edit

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Day 1b - Agent Architectures

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2. Summarizer Agent - Creates concise summaries from research findings

```
# Research Agent: Its job is to use the google_search tool and present findings.
research_agent = Agent(
    name="ResearchAgent",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    instruction="You are a specialized research agent. Your only job is to use the google_search tool to find 2-3 pieces of relevant information on the given topic and present it in a concise summary.",
    tools=[google_search],
    output_key="research_findings", # The result of this agent will be stored in the session
)

print("research_agent created.")
```

+ Code + Markdown

research_agent created.

Compartir Guardar versión 0

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

GOOGLE_API_KEY

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Day 1b - Agent Architectures

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```
# Summarizer Agent: Its job is to summarize the text it receives.
summarizer_agent = Agent(
    name="SummarizerAgent",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    # The instruction is modified to request a bulleted list for a clear output format.
    instruction="Read the provided research findings: {research_findings}
    Create a concise summary as a bulleted list with 3-5 key points.",
    output_key="final_summary",
)

print(✅ summarizer_agent created.)
```

✅ summarizer_agent created.

+ Code + Markdown

Refer to the ADK documentation for more information on [guiding agents with clear and specific instructions.](#)

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

GOOGLE_API_KEY

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Day 1b - Agent Architectures

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```
# Root Coordinator: Orchestrates the workflow by calling the sub-agents as tools.
root_agent = Agent(
    name="ResearchCoordinator",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    # This instruction tells the root agent HOW to use its tools (which are the other agents)
    instruction="You are a research coordinator. Your goal is to answer the user's query.
    1. First, you MUST call the 'ResearchAgent' tool to find relevant information on the topic.
    2. Next, after receiving the research findings, you MUST call the 'SummarizerAgent' tool to create a concise summary.
    3. Finally, present the final summary clearly to the user as your response.",
    # We wrap the sub-agents in 'AgentTool' to make them callable tools for the root agent.
    tools=[AgentTool(research_agent), AgentTool(summarizer_agent)],
)

print(✅ root_agent created.)
```

✅ root_agent created.

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

GOOGLE_API_KEY

+ Agregar secreto Copiar fragmento

https://www.kaggle.com/benchmarks

Humedad 09:56 a.m.

← → ↻ kaggle.com/code/sandovalaguirredavid/day-1b-agent-architectures/edit

Day 1b - Agent Architectures

Archivo Editar Vista Correr Ajustes Complementos Ayuda

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```
# Writer Agent: Writes the full blog post based on the outline from the previous agent.
writer_agent = Agent(
    name="WriterAgent",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    # The `{blog_outline}` placeholder automatically injects the state value from the previous agent.
    instruction="Following this outline strictly: {blog_outline}
    Write a brief, 200 to 300-word blog post with an engaging and informative tone.",
    output_key="blog_draft", # The result of this agent will be stored with this key.
)

print(✅ writer_agent created.)

✅ writer_agent created.
```

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecretsClient
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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Day 1b - Agent Architectures

Archivo Editar Vista Correr Ajustes Complementos Ayuda

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```
# Editor Agent: Edits and polishes the draft from the writer agent.
editor_agent = Agent(
    name="EditorAgent",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    # This agent receives the `{blog_draft}` from the writer agent's output.
    instruction="Edit this draft: {blog_draft}
    Your task is to polish the text by fixing any grammatical errors, improving the flow and readability.
    output_key='final_blog', # This is the final output of the entire pipeline.
)

print(✅ editor_agent created.)

✅ editor_agent created.
```

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecretsClient
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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Todos los favoritos

Day 1b - Agent Architectures

Archivo Editar Vista Correr Ajustes Complementos Ayuda

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```
print(✅ editor_agent created.)
```

✅ editor_agent created.

Then we bring the agents together under a sequential agent, which runs the agents in the order that they are listed:

```
root_agent = SequentialAgent(
    name="BlogPipeline",
    sub_agents=[outline_agent, writer_agent, editor_agent],
)

print(✅ Sequential Agent created.)
```

✅ Sequential Agent created.

+ Code + Markdown

Let's run the agent and give it a topic to write a blog post about:

Compartir Guardar versión 0

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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Todos los favoritos

Day 1b - Agent Architectures

Archivo Editar Vista Correr Ajustes Complementos Ayuda

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```
runner = InMemoryRunner(agent=root_agent)
response = await runner.run_debug(
    "Write a blog post about the benefits of multi-agent systems for software developers"
)
```

Created new session: debug_session_id

User > Write a blog post about the benefits of multi-agent systems for software developers

OutlineAgent > ### Blog Outline: Multi-Agent Systems for Software Developers

Headline: Unleash Superpowers: How Multi-Agent Systems Can Revolutionize Your Development Work flow

Introduction Hook: Are you tired of monolithic codebases and complex, tangled dependencies? Im agine a world where your software is built by a team of intelligent, specialized agents, each hand ling a specific task with unparalleled efficiency. This isn't science fiction; it's the power of m ulti-agent systems (MAS), and it's poised to transform how software developers work.

Main Sections:

1. The Power of Specialization: Breaking Down Complexity

Compartir Guardar versión 0

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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Day 1b - Agent Architectures

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4. **Aggregator Agent** - Combines all research findings into a single summary

```
# Tech Researcher: Focuses on AI and ML trends.
tech_researcher = Agent(
    name="TechResearcher",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    instruction="""Research the latest AI/ML trends. Include 3 key developments,
the main companies involved, and the potential impact. Keep the report very concise (100 \
tools=[google_search],
    output_key="tech_research", # The result of this agent will be stored in the session s
)

print("✅ tech_researcher created.")
```

✅ tech_researcher created.

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserS
user_secrets = UserSecretsClient
secret_value_0 = user_secrets.ge
```

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Day 1b - Agent Architectures

Archivo Editar Vista Correr Ajustes Complementos Ayuda

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```
# Health Researcher: Focuses on medical breakthroughs.
health_researcher = Agent(
    name="HealthResearcher",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    instruction="""Research recent medical breakthroughs. Include 3 significant advances,
their practical applications, and estimated timelines. Keep the report concise (100 words
tools=[google_search],
    output_key="health_research", # The result will be stored with this key.
)

print("✅ health_researcher created.")
```

✅ health_researcher created.

+ Code + Markdown

[]: # Finance Researcher: Focuses on fintech trends.

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_sec
```

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Day 1b - Agent Architectures

Archivo Editar Vista Correr Ajustes Complementos Ayuda

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```
# Finance Researcher: Focuses on fintech trends.
finance_researcher = Agent(
    name="FinanceResearcher",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    instruction="""Research current fintech trends. Include 3 key trends,
their market implications, and the future outlook. Keep the report concise (100 words).""",
    tools=[google_search],
    output_key="finance_research", # The result will be stored with this key.
)

print(✅ finance_researcher created.)
```

✅ finance_researcher created.

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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Day 1b - Agent Architectures

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```
),
# It uses placeholders to inject the outputs from the parallel agents, which are now in
instruction="""Combine these three research findings into a single executive summary:

**Technology Trends:**
{tech_research}

**Health Breakthroughs:**
{health_research}

**Finance Innovations:**
{finance_research}

Your summary should highlight common themes, surprising connections, and the most important
output_key="executive_summary", # This will be the final output of the entire system.
)

print(✅ aggregator_agent created.)
```

✅ aggregator_agent created.

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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Day 1b - Agent Architectures

Archivo Editar Vista Correr Ajustes Complementos Ayuda

Run All Code Sesión de borrador Cargando 0 m

```
# This agent runs ONCE at the beginning to create the first draft.
initial_writer_agent = Agent(
    name="InitialWriterAgent",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    instruction="""Based on the user's prompt, write the first draft of a short story (and
    Output only the story text, with no introduction or explanation.""",
    output_key="current_story", # Stores the first draft in the state.
)

print("✅ initial_writer_agent created.")
```

✅ initial_writer_agent created.

+ Code + Markdown

```
[ ]: # This agent's only job is to provide feedback or the approval signal. It has no tools.
critic_agent = Agent(
    name="CriticAgent",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    instruction="""You are a constructive story critic. Review the story provided below.
    Story: {current_story}

    Evaluate the story's plot, characters, and pacing.
    - If the story is well-written and complete, you MUST respond with the exact phrase:
    - Otherwise, provide 2-3 specific, actionable suggestions for improvement.""",
    output_key="critique", # Stores the feedback in the state.
)

print("✅ critic_agent created.")
```

✅ critic_agent created.

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

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Day 1b - Agent Architectures

Archivo Editar Vista Correr Ajustes Complementos Ayuda

Run All Code Sesión de borrador Cargando 0 m

```
# This agent's only job is to provide feedback or the approval signal. It has no tools.
critic_agent = Agent(
    name="CriticAgent",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    instruction="""You are a constructive story critic. Review the story provided below.
    Story: {current_story}

    Evaluate the story's plot, characters, and pacing.
    - If the story is well-written and complete, you MUST respond with the exact phrase:
    - Otherwise, provide 2-3 specific, actionable suggestions for improvement.""",
    output_key="critique", # Stores the feedback in the state.
)

print("✅ critic_agent created.")
```

✅ critic_agent created.

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

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Day 1b - Agent Architectures

Archivo Editar Vista Correr Ajustes Complementos Ayuda

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First, you'll define the `exit_loop` function:

```
# This is the function that the RefinerAgent will call to exit the loop.
def exit_loop():
    """Call this function ONLY when the critique is 'APPROVED', indicating the story is finished.
    """
    return {"status": "approved", "message": "Story approved. Exiting refinement loop."}

print('✅ exit_loop function created.')
```

✅ exit_loop function created.

+ Code + Markdown

To let an agent call this Python function, we wrap it in a `FunctionTool`. Then, we create a `RefinerAgent` that has this tool.

📌 **Notice its instructions:** this agent is the "brain" of the loop. It reads the `{critique}` from the `CriticAgent` and decides whether to (1) call the `exit_loop` tool or (2) rewrite the story.

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecretsClient
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

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Day 1b - Agent Architectures

Archivo Editar Vista Correr Ajustes Complementos Ayuda

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```
model=gemini({
    model="gemini-2.5-flash-lite",
    retry_options=retry_config
}),
instruction="""You are a story refiner. You have a story draft and critique.

Story Draft: {current_story}
Critique: {critique}

Your task is to analyze the critique.
- IF the critique is EXACTLY "APPROVED", you MUST call the 'exit_loop' function and no further action is needed.
- OTHERWISE, rewrite the story draft to fully incorporate the feedback from the critique.
output_key="current_story", # It overwrites the story with the new, refined version.
tools=[
    FunctionTool(exit_loop)
], # The tool is now correctly initialized with the function reference.

print('✅ refiner_agent created.')
```

✅ refiner_agent created.

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecretsClient
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

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Todos los favoritos

Day 1b - Agent Architectures

Archivo Editar Vista Correr Ajustes Complementos Ayuda

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↑ ↓ 🗑

```
# The LoopAgent contains the agents that will run repeatedly: Critic -> Refiner.
story_refinement_loop = LoopAgent(
    name="StoryRefinementLoop",
    sub_agents=[critic_agent, refiner_agent],
    max_iterations=2, # Prevents infinite loops
)

# The root agent is a SequentialAgent that defines the overall workflow: Initial Write -> R
root_agent = SequentialAgent(
    name="StoryPipeline",
    sub_agents=[initial_writer_agent, story_refinement_loop],
)

print(✅ Loop and Sequential Agents created.)
```

✅ Loop and Sequential Agents created.

+ Code + Markdown

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr

Copiar al portapapeles

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Todos los favoritos

Day 1b - Agent Architectures

Draft saved

File Edit View Run Settings Add-ons Help

+ ↻ 🔍 📄 📌 ▶ ▶▶ Run All Code Draft Session Loading12m)

↑ ↓ 🗑

```
runner = InMemoryRunner(agent=root_agent)
response = await runner.run_debug(
    "Write a short story about a lighthouse keeper who discovers a mysterious, glowing map"
)

### Created new session: debug_session_id

User > Write a short story about a lighthouse keeper who discovers a mysterious, glowing map
InitialWriterAgent > Elias, his face weathered like the sea-battered rocks, polished the great len
s of the lighthouse. For twenty years, the rhythmic sweep of light had been his only companion, sa
ve for the gulls' cries. Tonight, however, a different glow pulsed from the dusty seabed. A glint
caught his eye, something unusual washed ashore with the tide.

He scrambled down the winding stairs and out into the salt spray. There, nestled amongst kelp and
driftwood, lay a rolled parchment, impossibly preserved. Unfurling it, Elias gasped. It was a map,
but unlike any he'd ever seen. Lines of luminous, sapphire ink crisscrossed a parchment that shimm
ered with an internal light, depicting islands that danced and dissolved like sea mist. A single,
pulsing X marked a spot in the uncharted deep.

ClientError Traceback (most recent call last)
/tmp/ipykernel_49/2323074114.py in <cell line: 1>()
1 runner = InMemoryRunner(agent=root_agent)
```

Secrets

Code Snippet

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr

Copy to clipboard

GOOGLE_API_KEY

+ Add Secret Copy snippet

Dia 0

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kaggle.com/code/sandovalaguirredavid/day-0-troubleshooting-and-faqs/edit

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Todos los favoritos

Day 0 - Troubleshooting and FAQs

Borrador guardado

Compartir Guardar versión 0

Archivo Editar Vista Correr Ajustes Complementos Ayuda

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```
# Step 1: Import required modules and define retry configuration
from google.genai import types
from google.adk.models.google_llm import Gemini

retry_config = types.HttpRetryOptions(
    attempts=5, # Maximum number of retry attempts
    exp_base=7, # Exponential backoff multiplier (delay grows: 1s, 7s, 49s...)
    initial_delay=1, # Initial delay before first retry (in seconds)
    http_status_codes=[429, 500, 503, 504], # Retry on these HTTP errors
)

print('✅ Retry configuration created')
```

✅ Retry configuration created

+ Code + Markdown

Step 2: Apply retry configuration to your model

When creating your agent, replace the simple model string with a `Gemini` object that includes retry options:

```
# Example: Complete agent with retry configuration
from google.adk.agents import LlmAgent

agent = LlmAgent(
    name="my_agent",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config, # Apply retry config here
    ),
    instruction="You are a helpful assistant.",
)

print('✅ Agent created with automatic retry on HTTP errors')
```

✅ Agent created with automatic retry on HTTP errors

+ Code + Markdown

5. InvalidArgument or invalid API key

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

☒ GOOGLE_API_KEY *****

+ Agregar secreto Copiar fragmento

Usar el chat en NotebookLM

Importante: Actualmente, esta función puede estar limitada en la aplicación móvil NotebookLM. [Más información sobre las funciones compatibles de la aplicación móvil NotebookLM](#)

Después de subir tus fuentes, puedes hacer lo siguiente:

- Hazle preguntas al modelo sobre el material de tu fuente.
- Da instrucciones al modelo para que realice acciones.

NotebookLM utiliza citas directas, texto e imágenes de tus fuentes como citas para responder a tus preguntas y realizar acciones. Estas citas te ayudan a comprobar la precisión de la respuesta. Puedes colocar el cursor sobre cualquier cita para ver el texto completo citado inmediatamente. Si seleccionas una cita, NotebookLM se desplazará automáticamente hasta la ubicación de la cita para que puedas verla fácilmente en contexto.

En tu cuaderno, puedes usar la casilla de cada fuente para incluir o excluir determinadas

Ayuda

- Información sobre NotebookLM
- Usar el chat en NotebookLM
- Crear un cuaderno en NotebookLM
- Añadir o descubrir nuevas fuentes para tu cuaderno
- Crear y añadir notas en NotebookLM
- Usar mapas conceptuales en NotebookLM
- Generar un resumen de audio en NotebookLM
- Generar un resumen de video en NotebookLM



The Google Intelligence Ecosystem

+ Crear cuaderno

Compartir

Configuración

Fuentes

+ Agregar fuentes

Buscar fuentes nuevas en la Web

Web Fast Research

Seleccionar todo

Texto pegado

Chat



The Google Intelligence Ecosystem

1 fuente · 30 abr 2026

El texto describe el **ecosistema de inteligencia artificial** desarrollado por Google, destacando herramientas clave diseñadas para diferentes propósitos. **Gemini** se posiciona como el motor principal de lenguaje capaz de generar contenido y analizar datos, funcionando como el núcleo de esta tecnología. Por otro lado, el **Asistente de Google** tradicional se enfoca en tareas de voz y control

Comienza a escribir... 1 fuente

Studio

Resumen en... Presentación...

Resumen en... Mapa mental

Informes Tarjetas...

Cuestionario Infografía

Tabla de datos

Los datos de salida de Studio se guardarán

Agregar nota

Es posible que NotebookLM muestre información imprecisa. Verifica las respuestas.

youtube.com/live/ZaUcqznlhv8

YouTube MX

Buscar

Kanchana Patlolla

Anant Nawalgaria

DAY 1 Livestream - 5-Day AI Agents Intensive Course

Kaggle

5-Day AI Agents Intensive Course with Google

50/10 Pomodoro Timer with Brown Noise

Colo bergel tapachula mexico

Crabb/Benesh vs. Cottafava/Dal Corso - Pool Play Highlights

dopamina - Peso Pluma & Tito Double P (Lyric Video)

kaggle.com/learn-guide/5-day-agents

kaggle

Buscar

Iniciar sesión

Registro

Tareas del Día 1

1. Completa la Unidad 1 – "Introducción a los agentes":

- Escucha el episodio del podcast que resume esta unidad.
- Para complementar el podcast, lee el documento técnico "Introducción a los agentes".
- Completa estos codelabs en Kaggle:
 - Crea tu primer agente usando Gemini y ADK.
 - Crea tus primeros sistemas multiagente con ADK.
 - Asegúrate de verificar tu cuenta de Kaggle por teléfono antes de empezar, es necesario para los codelabs.
 - También contamos con una guía de solución de problemas para los laboratorios de programación. No olvides consultarla para encontrar soluciones a los problemas más comunes.
 - ¿Quieres tener una conversación interactiva? Prueba a añadir los documentos técnicos a NotebookLM.

[Opcional]: Mira la retransmisión en directo grabada en YouTube. Kanchana Patlolla y Anant Nawalgaria presentaron la primera retransmisión en directo en el canal de YouTube de Kaggle. Se les unieron los autores del codelab, Kristopher Overholt y Hangfei Lin, junto con invitados especiales de Google: Alan Blount, Mike Clark, Michael Gerstenhaber y Antonio Gulli, para hablar sobre las tareas y compartir ideas.

Dia 2 A

The screenshot shows a YouTube video player interface. The video title is "Whitepaper Companion Podcast - Agent Tools & Interoperability with MCP" by Kaggle. The video has 5.5K likes and 39K views. The video player is displaying a thumbnail for the podcast, which features the text "Agent Tools & Interoperability with MCP" and the authors: Mike Styer, Kanchana Patlolla, Madhuranjan Mohan, and Sal Diaz. The video is part of a series called "Whitepaper Companion Podcast".

Below the video player, the Kaggle website is visible. The page title is "Herramientas para agentes e interoperabilidad con MCP". The authors listed are Mike Styer, Kanchana Patlolla, Madhuranjan Mohan, and Sal Diaz. The page includes an introduction section and a link to the technical report.

The introduction section states: "En este documento, primero analizamos la naturaleza de las herramientas utilizadas por los modelos base: qué son y cómo utilizarlas. Presentamos algunas buenas prácticas y directrices para diseñar y utilizar herramientas eficaces. A continuación, examinamos el Protocolo de Contexto del Modelo (MCP), describiendo sus componentes básicos y algunos de los retos y riesgos que conlleva. Finalmente, analizamos en profundidad los desafíos de seguridad que plantea el MCP al implementarse en un entorno empresarial y conectarse a sistemas externos de alto valor."

The page also includes a section titled "Lea el informe técnico a continuación."

The bottom of the screenshot shows the Windows taskbar with the date 30/04/2026 and the time 10:21 a.m.

The image shows a web browser window displaying a Kaggle notebook titled "Day 2a - Agent Tools". The notebook is in "Code" view and shows a series of Python import statements for Google ADK components, followed by a print statement confirming successful imports. The output shows a green checkmark and the text "ADK components imported successfully.". On the right sidebar, there is a section titled "Misterios" with a "Fragmento de código" (Code snippet) containing a snippet of Python code for handling secrets. Below this is a "Copiar al portapapeles" (Copy to clipboard) button. At the bottom of the notebook, there is a section titled "1.4: Helper functions" with a description: "Helper function that prints the generated Python code and results from the code execution tool:". The browser's address bar shows the URL "kaggle.com/code/sandovalaguirredavid/day-2a-agent-tools/edit".

← → ↺

kaggle.com/code/sandovalguirredavid/day-2a-agent-tools/edit

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📁 Todos los favoritos

☰

Day 2a - Agent Tools

Archivo Editar Vista Correr Ajustes Complementos Ayuda

+ - ✂️ 📄 📁 ▶️ ⏮️ ⏭️ Run All Code Sesión de borrador Cargando(0 m)

retry_config = types.HttpRetryOptions

When working with LLMs, you may encounter transient errors like rate limits or temporary service unavailability. Retry options automatically handle these failures by retrying the request with exponential backoff.

+ Code + Markdown

▶️

```
retry_config = types.HttpRetryOptions(
    attempts=5, # Maximum retry attempts
    exp_base=7, # Delay multiplier
    initial_delay=1,
    http_status_codes=[429, 500, 503, 504], # Retry on these HTTP errors
)
```

+ Code + Markdown

🏷️

Section 2: What are Custom Tools?

Custom Tools are tools you build yourself using your own code and business logic. Unlike built-in tools that come ready-made with ADK, custom tools give you complete control over functionality.

When to use Custom Tools?

🔗 Compartir

🔄 Guardar versión 0

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecretsClient
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

📄 Copiar al portapapeles

☒ GOOGLE_API_KEY *****

+ Agregar secreto 📄 Copiar fragmento

The screenshot displays the Kaggle code editor for a notebook titled "Day 2a - Agent Tools". The editor interface includes a top navigation bar with options like "Archivo", "Editar", "Vista", "Correr", "Ajustes", "Complementos", and "Ayuda". A right sidebar shows "Misterios" and "Fragmento de código". The main code area contains a Python function `get\_exchange\_rate` that interacts with a database to retrieve exchange rates. The code output shows a successful test for USD to EUR conversion.

```
base = base_currency.lower()
target = target_currency.lower()

# Return structured result with status
rate = rate_database.get(base, {}).get(target)
if rate is not None:
    return {"status": "success", "rate": rate}
else:
    return {
        "status": "error",
        "error_message": f"Unsupported currency pair: {base_currency}/{target_currency}"
    }

print("✅ Exchange rate function created")
print(f"🧪 Test: {get_exchange_rate('USD', 'EUR')}")
```

The output of the code execution is shown below the code block:

```
✅ Exchange rate function created
🧪 Test: {'status': 'success', 'rate': 0.93}
```

← → ↺

kaggle.com/code/sandovalaguirredavid/day-2a-agent-tools/edit

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Day 2a - Agent Tools

Borrador guardado

Archivo Editar Vista Correr Ajustes Complementos Ayuda

Compartir Guardar versión 0

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1. Check the status of the tool in each tool's response for errors.

2. Check the status of the tool in each tool's response for errors.

3. Check the status of the tool in each tool's response for errors.

4. Calculate the final amount after fees based on the output from 'get_fee_for_payment_method'.

5. First, state the final converted amount.

Then, explain how you got that result by showing the intermediate amounts. Your explanation should include the original value in the original currency, the amount remaining after the fee, and the exchange rate.

If any tool returns status "error", explain the issue to the user clearly.

"""

tools=[get_fee_for_payment_method, get_exchange_rate],

)

print("✅ Currency agent created with custom function tools")

print("🔧 Available tools:")

print(" • get_fee_for_payment_method - Looks up company fee structure")

print(" • get_exchange_rate - Gets current exchange rates")

"""

✅ Currency agent created with custom function tools

🔧 Available tools:

• get_fee_for_payment_method - Looks up company fee structure

• get_exchange_rate - Gets current exchange rates

+ Code + Markdown

Misterios

Fragmento de código

from kaggle_secrets import UserSecretsClient
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0

Copiar al portapapeles

GOOGLE_API_KEY

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Todos los favoritos

Day 2a - Agent Tools

Borrador guardado

Archivo Editar Vista Correr Ajustes Complementos Ayuda

Compartir Guardar versión 0

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1. Check the status of the tool in each tool's response for errors.

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4. Calculate the final amount after fees based on the output from 'get_fee_for_payment_method'.

5. First, state the final converted amount.

Then, explain how you got that result by showing the intermediate amounts. Your explanation should include the original value in the original currency, the amount remaining after the fee, and the exchange rate.

If any tool returns status "error", explain the issue to the user clearly.

"""

tools=[get_fee_for_payment_method, get_exchange_rate],

)

print("✅ Currency agent created with custom function tools")

print("🔧 Available tools:")

print(" • get_fee_for_payment_method - Looks up company fee structure")

print(" • get_exchange_rate - Gets current exchange rates")

"""

✅ Currency agent created with custom function tools

🔧 Available tools:

• get_fee_for_payment_method - Looks up company fee structure

• get_exchange_rate - Gets current exchange rates

+ Code + Markdown

Misterios

Fragmento de código

from kaggle_secrets import UserSecretsClient
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0

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Day 2a - Agent Tools

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```
calculation_agent = LlmAgent(
    name="CalculationAgent",
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    instruction="""You are a specialized calculator that ONLY responds with Python code.

    Your task is to take a request for a calculation and translate it into a single block

    **RULES:**
    1. Your output MUST be ONLY a Python code block.
    2. Do NOT write any text before or after the code block.
    3. The Python code MUST calculate the result.
    4. The Python code MUST print the final result to stdout.
    5. You are PROHIBITED from performing the calculation yourself. Your only job is to

    Failure to follow these rules will result in an error.
    """
)

code_executor=BuiltInCodeExecutor(), # Use the built-in Code Executor Tool. This gives
```

2. Update the Agent's instruction and toolset

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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Day 2a - Agent Tools

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```
""" * The exchange rate applied.

tools=[
    get_fee_for_payment_method,
    get_exchange_rate,
    AgentTool(agent=calculation_agent), # Using another agent as a tool!
],
)

print(✅ Enhanced currency agent created")
print(🔗 New capability: Delegates calculations to specialist agent")
print(🔗 Tool types used:")
print(" • Function Tools (fees, rates)")
print(" • Agent Tool (calculation specialist)")

✅ Enhanced currency agent created
🔗 New capability: Delegates calculations to specialist agent
🔗 Tool types used:
• Function Tools (fees, rates)
• Agent Tool (calculation specialist)
```

3. Create a new agent to handle currency calculations

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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[15]:
Define a runner
enhanced_runner = InMemoryRunner(agent=enhanced_currency_agent)

Test the enhanced agent
response = await enhanced_runner.run_debug(
 "Convert 1,250 USD to INR using a Bank Transfer. Show me the precise calculation."
)

Created new session: debug_session_id

User > Convert 1,250 USD to INR using a Bank Transfer. Show me the precise calculation.

ClientError Traceback (most recent call last)
/tmp/ipykernel_49/433477422.py in <cell line: 1>()
1 # Test the enhanced agent
----> 2 response = await enhanced_runner.run_debug(
3 "Convert 1,250 USD to INR using a Bank Transfer. Show me the precise calculation."
4)

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0

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Day 2a - Agent Tools

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show_python_code_and_result(response)

NameError Traceback (most recent call last)
/tmp/ipykernel_49/3237811916.py in <cell line: 0>()
----> 1 show_python_code_and_result(response)

NameError: name 'response' is not defined

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0

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🙄 3.3: Agent Tools vs Sub-Agents: What's the Difference?

This is a common question! Both involve using multiple agents, but they work very differently:

Agent Tools (what we're using):

- Agent A calls Agent B as a tool
- Agent B's response goes back to Agent A

Dia 2 B

The image displays two screenshots of the Kaggle code editor interface, showing the setup of a Google API key for Day 2b of the Kaggle 5-day Agents course.

Top Screenshot:

- Page Title:** Day 2b - Agent Tools Best Practices
- URL:** kaggle.com/code/sandovalaguirredavid/day-2b-agent-tools-best-practices/edit
- Toolbar:** Archivo, Editar, Vista, Correr, Ajustes, Complementos, Ayuda. Botones: Ejecutar todo, Code, Sesión de borrador, Cargando 0m.
- Code Editor:** Contains a license notice for Apache License, Version 2.0.
- Right Panel (Misterios):** Shows a code snippet for importing secrets from Kaggle and a section for adding secrets, including a checked checkbox for `GOOGLE_API_KEY`.

Bottom Screenshot:

- Page Title:** Day 2b - Agent Tools Best Practices
- URL:** kaggle.com/code/sandovalaguirredavid/day-2b-agent-tools-best-practices/edit
- Toolbar:** Archivo, Editar, Vista, Correr, Ajustes, Complementos, Ayuda. Botones: Ejecutar todo, Code, Sesión de borrador, Cargando 0m.
- Code Editor:** Contains Python code for importing `os` and `UserSecretsClient` from `kaggle_secrets`, and setting up the environment with the `GOOGLE_API_KEY`.
- Right Panel (Misterios):** Shows the same code snippet and secret management section as the top screenshot.

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Day 2b - Agent Tools Best Practices

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```
import uuid
from google.genai import types

from google.adk.agents import LlmAgent
from google.adk.models.google_llm import Gemini
from google.adk.runners import Runner
from google.adk.sessions import InMemorySessionService

from google.adk.tools.mcp_tool.mcp_toolset import McpToolset
from google.adk.tools.tool_context import ToolContext
from google.adk.tools.mcp_tool.mcp_session_manager import StdioConnectionParams
from mcp import StdioServerParameters

from google.adk.apps.app import App, ResumabilityConfig
from google.adk.tools.function_tool import FunctionTool

print("✅ ADK components imported successfully.")
```

✅ ADK components imported successfully.

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

☒ GOOGLE_API_KEY *****

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Day 2b - Agent Tools Best Practices

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```
retry_config = types.HttpRetryOptions(
    attempts=5, # Maximum retry attempts
    exp_base=7, # Delay multiplier
    initial_delay=1,
    http_status_codes=[429, 500, 503, 504], # Retry on these HTTP errors
)
```

+ Code + Markdown

Section 2: Model Context Protocol

So far, you have learned how to create custom functions for your agents. But connecting to external systems (GitHub, databases, Slack) requires writing and maintaining API clients.

Model Context Protocol (MCP) is an open standard that lets agents use community-built integrations. Instead of writing your own integrations and API clients, just connect to an existing MCP server.

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

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Day 2b - Agent Tools Best Practices

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```
# MCP integration with Everything Server
mcp_image_server = McpToolset(
    connection_params=StdioConnectionParams(
        server_params=StdioServerParameters(
            command="npx", # Run MCP server via npx
            args=[
                "-y", # Argument for npx to auto-confirm install
                "@modelcontextprotocol/server-everything",
            ],
            tool_filter=["getTinyImage"],
        ),
        timeout=30,
    )
)

print("✅ MCP Tool created")
```

✅ MCP Tool created

+ Code + Markdown

Behind the scenes:

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Compartir Guardar versión 0

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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Day 2b - Agent Tools Best Practices

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```
# Create image agent with MCP integration
image_agent = LlmAgent(
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    name="image_agent",
    instruction="Use the MCP Tool to generate images for user queries",
    tools=[mcp_image_server],
)

[ ]: from google.adk.runners import InMemoryRunner
runner = InMemoryRunner(agent=image_agent)
```

Create the runner:

[]: from google.adk.runners import InMemoryRunner
runner = InMemoryRunner(agent=image_agent)

Compartir Guardar versión 0

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

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Day 2b - Agent Tools Best Practices

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Display the image:

The server returns base64-encoded image data. Let's decode and display it:

```
from IPython.display import display, Image as IPIImage
import base64

for event in response:
    if event.content and event.content.parts:
        for part in event.content.parts:
            if hasattr(part, "function_response") and part.function_response:
                for item in part.function_response.response.get("content", []):
                    if item.get("type") == "image":
                        display(IPIImage(data=base64.b64decode(item["data"])))
```

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr
```

Copiar al portapapeles

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Day 2b - Agent Tools Best Practices

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Borrador guardado

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```
# SCENARIO 3: The tool is called AGAIN and is now resuming. Handle approval response -
if tool_context.tool_confirmation.confirmed:
    return {
        "status": "approved",
        "order_id": f"ORD-{num_containers}-HUMAN",
        "num_containers": num_containers,
        "destination": destination,
        "message": f"Order approved: {num_containers} containers to {destination}",
    }
else:
    return {
        "status": "rejected",
        "message": f"Order rejected: {num_containers} containers to {destination}",
    }

print("✅ Long-running functions created!")
```

Long-running functions created!

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr
```

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Day 2b - Agent Tools Best Practices

Borrador guardado

Compartir Guardar versión 0

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Run All

Code

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```
# Create shipping agent with pausable tool
shipping_agent = LLMAgent(
    name="shipping_agent",
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    instruction="""You are a shipping coordinator assistant.

When users request to ship containers:
1. Use the place_shipping_order tool with the number of containers and destination
2. If the order status is 'pending', inform the user that approval is required
3. After receiving the final result, provide a clear summary including:
   - Order status (approved/rejected)
   - Order ID (if available)
   - Number of containers and destination
4. Keep responses concise but informative
"""
    tools=[FunctionTool(func=place_shipping_order)],
)

print("✅ Shipping Agent created!")
```

✅ Shipping Agent created!

+ Code + Markdown

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr

Copiar al portapapeles

GOOGLE_API_KEY

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Day 2b - Agent Tools Best Practices

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Run All

Code

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```
# Wrap the agent in a resumable app - THIS IS THE KEY FOR LONG-RUNNING OPERATIONS!
shipping_app = App(
    name="shipping_coordinator",
    root_agent=shipping_agent,
    resumability_config=ResumabilityConfig(is_resumable=True),
)

print("✅ Resumable app created!")
```

✅ Resumable app created!

/tmp/ipynkernel_49/3673777575.py:5: UserWarning: [EXPERIMENTAL] ResumabilityConfig: This feature is experimental and may change or be removed in future versions without notice. It may introduce breaking changes at any time.

resumability_config=ResumabilityConfig(is_resumable=True),

+ Code + Markdown

Step 3: Create Session and Runner with the App

ResumabilityConfig is experimental. Instead of ResumabilityConfig, use the resumable flag on the agent.

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr

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```
session_service = InMemorySessionService()

# Create runner with the resumable app
shipping_runner = Runner(
    app=shipping_app, # Pass the app instead of the agent
    session_service=session_service,
)

print('✅ Runner created!')
```

✅ Runner created!

+ Code + Markdown

✅ Recap: Your pausable shipping agent is now complete!

You've created:

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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↑ ↓ 🗑

```
def check_for_approval(events):
    """Check if events contain an approval request.

    Returns:
        dict with approval details or None
    """
    for event in events:
        if event.content and event.content.parts:
            for part in event.content.parts:
                if (
                    part.function_call
                    and part.function_call.name == "adk_request_confirmation"
                ):
                    return {
                        "approval_id": part.function_call.id,
                        "invocation_id": event.invocation_id,
                    }
    return None
```

+ Code + Markdown

print_agent_response() - Displays agent text

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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```
return None
```

`print_agent_response()` - Displays agent text

- Simple helper to extract and print text from events

```
def print_agent_response(events):  
    """Print agent's text responses from events."""  
    for event in events:  
        if event.content and event.content.parts:  
            for part in event.content.parts:  
                if part.text:  
                    print(f"Agent > {part.text}")
```

+ Code + Markdown

`create_approval_response()` - Formats the human decision

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Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets  
user_secrets = UserSecretsClient()  
secret_value_0 = user_secrets.get_secret_value_0
```

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Day 2b - Agent Tools Best Practices

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- Takes the approval info and boolean decision (True/False) from the human
- Creates a `FunctionResponse` that ADK understands
- Wraps it in a `Content` object to send back to the agent

+ Code + Markdown

```
[21]:  
def create_approval_response(approval_info, approved):  
    """Create approval response message."""  
    confirmation_response = types.FunctionResponse(  
        id=approval_info["approval_id"],  
        name="adk_request_confirmation",  
        response={"confirmed": approved},  
    )  
    return types.Content(  
        role="user", parts=[types.Part(function_response=confirmation_response)]  
    )  
  
print("✅ Helper functions defined")
```

✅ Helper functions defined

Compartir Guardar versión 0

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets  
user_secrets = UserSecretsClient()  
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

☒ GOOGLE_API_KEY *****

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5-Day AI Agents Intensive Course

Whitepaper Companion Podcast

Context Engineering: Sessions & Memory

Authors: Kimberly Milam and Antonio Gulli

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Ingeniería de contexto: sesiones y memoria

Autores: Kimberly Milam y Antonio Gulli

Introducción

Este documento técnico explora el papel fundamental de las sesiones y la memoria en la creación de agentes LLM inteligentes y con estado, para que los desarrolladores puedan crear experiencias de IA más potentes, personalizadas y persistentes. Para que los modelos de lenguaje a gran escala (LLM) recuerden, aprendan y personalicen las interacciones, los desarrolladores deben ensamblar y gestionar información de forma dinámica dentro de su ventana de contexto, un proceso conocido como ingeniería de contexto.

Lea el informe técnico a continuación.

DIA 3 A

The image displays two screenshots of the Kaggle Notebook Editor interface, showing the setup of the Gemini API key and the import of ADK components.

Top Screenshot:

- Page Title:** Day 3a - Agent Sessions
- Navigation:** Archivo, Editar, Vista, Correr, Ajustes, Complementos, Ayuda
- Code Cell:** The code cell is titled "1.3: Import ADK components". It contains the following Python code:

```
import os
from kaggle_secrets import UserSecretsClient

try:
    GOOGLE_API_KEY = UserSecretsClient().get_secret("GOOGLE_API_KEY")
    os.environ["GOOGLE_API_KEY"] = GOOGLE_API_KEY
    print("✅ Gemini API key setup complete.")
except Exception as e:
    print(f"🚨 Authentication Error: Please make sure you have added 'GOOGLE_API_KEY' to yo")
```

The output of the code cell shows a green checkmark and the message "Gemini API key setup complete."
- Right Panel:** The "Misterios" panel shows a "Fragmento de código" (Code Snippet) with the same code as the cell. Below it, there is a "Copiar al portapapeles" (Copy to clipboard) button and a "GOOGLE_API_KEY" secret value.

Bottom Screenshot:

- Page Title:** Day 3a - Agent Sessions
- Navigation:** Archivo, Editar, Vista, Correr, Ajustes, Complementos, Ayuda
- Code Cell:** The code cell is titled "1.4: Helper functions". It contains the following Python code:

```
from typing import Any, Dict

from google.adk.agents import Agent, LlmAgent
from google.adk.apps.app import App, EventsCompactionConfig
from google.adk.models.google_llm import Gemini
from google.adk.sessions import DatabaseSessionService
from google.adk.sessions import InMemorySessionService
from google.adk.runners import Runner
from google.adk.tools.tool_context import ToolContext
from google.genai import types

print("✅ ADK components imported successfully.")
```

The output of the code cell shows a green checkmark and the message "ADK components imported successfully."
- Right Panel:** The "Misterios" panel shows a "Fragmento de código" (Code Snippet) with the same code as the cell. Below it, there is a "Copiar al portapapeles" (Copy to clipboard) button and a "GOOGLE_API_KEY" secret value.

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query = types.Content(role= user , parts=[types.Part(text=query)])

Stream the agent's response asynchronously

async for event in runner_instance.run_async(

user_id=USER_ID, session_id=session.id, new_message=query

):

Check if the event contains valid content

if event.content and event.content.parts:

Filter out empty or "None" responses before printing

if (

event.content.parts[0].text != "None"

and event.content.parts[0].text

):

print(f"(MODEL_NAME) > ", event.content.parts[0].text)

else:

print("No queries!")

print(✅ Helper functions defined.)

Helper functions defined.

+ Code + Markdown

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets

user_secrets = UserSecretsClient()

secret_value_0 = user_secrets.get_secret_value_0

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when working with LLMs, you may encounter transient errors like rate limits or temporary service unavailability. retry options automatically handle these failures by retrying the request with exponential backoff.

retry_config = types.HttpRetryOptions(

attempts=5, # Maximum retry attempts

exp_base=7, # Delay multiplier

initial_delay=1,

http_status_codes=[429, 500, 503, 504], # Retry on these HTTP errors

)

+ Code + Markdown

Section 2: Session Management

2.1 The Problem

At their core, Large Language Models are **inherently stateless**. Their awareness is confined to the information you provide in a single API call. This means an agent without proper context management will reset to the current prompt.

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets

user_secrets = UserSecretsClient()

secret_value_0 = user_secrets.get_secret_value_0

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```
# Step 2: Set up Session Management
# InMemorySessionService stores conversations in RAM (temporary)
session_service = InMemorySessionService()

# Step 3: Create the Runner
runner = Runner(agent=root_agent, app_name=APP_NAME, session_service=session_service)

print("✅ Stateful agent initialized!")
print(f"  - Application: {APP_NAME}")
print(f"  - User: {USER_ID}")
print(f"  - Using: {session_service.__class__.__name__}")

✅ Stateful agent initialized!
  - Application: default
  - User: default
  - Using: InMemorySessionService
```

+ Code + Markdown

2.5 Testing Our Stateful Agent

Now let's see the magic of sessions in action!

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Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr

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```
# Run a conversation with two queries in the same session
# Notice: Both queries are part of the SAME session, so context is maintained
await run_session(
    runner,
    [
        "Hi, I am Sam! What is the capital of United States?",
        "Hello! What is my name?", # This time, the agent should remember!
    ],
    "stateful-agentic-session",
)

### Session: stateful-agentic-session

User > Hi, I am Sam! What is the capital of United States?
gemini-2.5-flash-lite > Hi Sam! The capital of the United States is Washington, D.C.

User > Hello! What is my name?
gemini-2.5-flash-lite > Your name is Sam.
```

+ Code + Markdown

🔥 Success! The agent remembered your name because both queries were part of the same session. The Runner automatically maintained the conversation history.

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr

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conversation.

```
# Run this cell after restarting the kernel. All this history will be gone...
await run_session(
    runner,
    ["What did I ask you about earlier?", "And remind me, what's my name?"],
    "stateful-agentic-session",
) # Note, we are using same session name

### Session: stateful-agentic-session

User > What did I ask you about earlier?
gemini-2.5-flash-lite > You asked me about the capital of the United States.

User > And remind me, what's my name?
gemini-2.5-flash-lite > Your name is Sam.
```

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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```
chatbot_agent = LLMagent(
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    name="text_chat_bot",
    description="A text chatbot with persistent memory",
)

# Step 2: Switch to DatabaseSessionService
# SQLite database will be created automatically
db_url = "sqlite:///my_agent_data.db" # Local SQLite file
session_service = DatabaseSessionService(db_url=db_url)

# Step 3: Create a new runner with persistent storage
runner = Runner(agent=chatbot_agent, app_name=APP_NAME, session_service=session_service)

print("✅ Upgraded to persistent sessions!")
print(f"  - Database: my_agent_data.db")
print(f"  - Sessions will survive restarts!")
```

✅ Upgraded to persistent sessions!
- Database: my_agent_data.db
- Sessions will survive restarts!

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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After the conversation, we'll inspect the `my_agent_data.db` SQLite database directly to see how the conversation events (the user queries and model responses) are stored.

```
await run_session(
    runner,
    ["Hi, I am Sam! What is the capital of the United States?", "Hello! What is my name?"
    "test-db-session-01",
])
```

Session: test-db-session-01

User > Hi, I am Sam! What is the capital of the United States?
gemini-2.5-flash-lite > Hi Sam! The capital of the United States is Washington, D.C.

User > Hello! What is my name?
gemini-2.5-flash-lite > Your name is Sam.

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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agent should still remember that our name is Sam from the first test run. This demonstrates the power of persistent sessions.

```
await run_session(
    runner,
    ["What is the capital of India?", "Hello! What is my name?"
    "test-db-session-01",
])
```

Session: test-db-session-01

User > What is the capital of India?
gemini-2.5-flash-lite > The capital of India is New Delhi.

User > Hello! What is my name?
gemini-2.5-flash-lite > Your name is Sam.

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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3.5 Let's verify that the session data is isolated

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```
print("\n✅ SUCCESS! Found the Compaction Event:")
print(f"  Author: {event.author}")
print(f"\n Compacted information: {event}")
found_summary = True
break

if not found_summary:
    print(
        "\n❌ No compaction event found. Try increasing the number of turns in the demo."
    )

--- Searching for Compaction Summary Event ---

✅ SUCCESS! Found the Compaction Event:
  Author: user

Compacted information: model_version=None content=None grounding_metadata=None partial=None turn_
complete=None finish_reason=None error_code=None error_message=None interrupted=None custom_metada
ta=None usage_metadata=None live_session_resumption_update=None input_transcription=None output_tr
anscription=None avg_logprobs=None logprobs_result=None cache_metadata=None citation_metadata=None
invocation_id='2bb390a8-190b-4ffa-8662-da3f0bfc74fd' author='user' actions=EventActions(skip_summa
rization=None, state_delta={}, artifact_delta={}, transfer_to_agent=None, escalate=None, requested
_auth_configs={}, requested_tool_confirmations={}, compaction={'start_timestamp': 1777961753.27760
```

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr
```

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```
tool_context.state["user:name"] = user_name
tool_context.state["user:country"] = country

return {"status": "success"}

# This demonstrates how tools can read from session state.
def retrieve_userinfo(tool_context: ToolContext) -> Dict[str, Any]:
    """
    Tool to retrieve user name and country from session state.
    """
    # Read from session state
    user_name = tool_context.state.get("user:name", "Username not found")
    country = tool_context.state.get("user:country", "Country not found")

    return {"status": "success", "user_name": user_name, "country": country}

print("✅ Tools created.")

✅ Tools created.
```

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr
```

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Retrieve the session and inspect its state
session = await session_service.get_session(
 app_name=APP_NAME, user_id=USER_ID, session_id="state-demo-session"
)

print("Session State Contents:")
print(session.state)
print("\n Notice the 'user:name' and 'user:country' keys storing our data!")

Session State Contents:
{'user:name': 'Sam', 'user:country': 'Poland'}

 Notice the 'user:name' and 'user:country' keys storing our data!

+ Code + Markdown

5.5 Session State Isolation

As we've already seen, an important characteristic of session state is that it's isolated per session. Let's demonstrate this by starting a new session:

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Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr

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Start a completely new session - the agent won't know our name
await run_session(
 runner,
 ["Hi there, how are you doing today? What is my name?"],
 "new-isolated-session",
)

Expected: The agent won't know the name because this is a different session

Session: new-isolated-session

User > Hi there, how are you doing today? What is my name?
gemini-2.5-flash-lite > Hello! I'm doing great. I'm sorry, I don't know your name yet. Is there a
nything I can help you with?

+ Code + Markdown

5.6 Cross-Session State Sharing

While sessions are isolated by default, you might notice something interesting. Let's check the state of our new session (new-isolated-session):

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr

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Day 3a - Agent Sessions

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While sessions are isolated by default, you might notice something interesting. Let's check the state of our new session (new-isolated-session):

```
# Check the state of the new session
session = await session_service.get_session(
    app_name=APP_NAME, user_id=USER_ID, session_id="new-isolated-session"
)

print("New Session State:")
print(session.state)

# Note: Depending on implementation, you might see shared state here.
# This is where the distinction between session-specific and user-specific state becomes im
```

New Session State:
{'user:name': 'Sam', 'user:country': 'Poland'}

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr
```

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Cleanup

```
# Clean up any existing database to start fresh (if Notebook is restarted)
import os

if os.path.exists("my_agent_data.db"):
    os.remove("my_agent_data.db")
print("✅ Cleaned up old database files")
```

✅ Cleaned up old database files

+ Code + Markdown

Summary

🎉 Congratulations! You've learned the fundamentals of building stateful AI agents:

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr
```

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▶▶

Run All

Code

⋮

✔ Congratulations! You did it🎉

Note: No submission required!

This notebook is for your hands-on practice and learning only. You **do not** need to submit it anywhere to complete the course.

Learn More

Refer to the following documentation to learn more:

- ADK Documentation
- ADK Sessions
- ADK Session-State
- ADK Session Compaction

Next Steps - Long Term Memory Systems (Part 2)

Why do we need memory?

In this notebook, we manually identified a couple characteristic (username and country) and built tools to manage it. But

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Misterios

Fragmento de código

from kaggle_secrets import UserSecretsClient\nuser_secrets = UserSecretsClient()\nsecret_value_0 = user_secrets.get_secret_value_0

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Ejecutar todo

Code

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↑ ↓

```
import os
from kaggle_secrets import UserSecretsClient

try:
    GOOGLE_API_KEY = UserSecretsClient().get_secret("GOOGLE_API_KEY")
    os.environ["GOOGLE_API_KEY"] = GOOGLE_API_KEY
    print("✅ Gemini API key setup complete.")
except Exception as e:
    print(f"🚨 Authentication Error: Please make sure you have added 'GOOGLE_API_KEY' to yo
    )

✅ Gemini API key setup complete.
```

+ Code + Markdown

1.3: Import ADK components

Now, import the specific components you'll need from the Agent Development Kit and the Generative AI library. This keeps your code organized and ensures we have access to the necessary building blocks.

Misterios

Fragmento de código

from kaggle_secrets import UserSecretsClient
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0

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Code

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↑ ↓

```
from google.adk.agents import LlmAgent
from google.adk.models.google_llm import Gemini
from google.adk.runners import Runner
from google.adk.sessions import InMemorySessionService
from google.adk.memory import InMemoryMemoryService
from google.adk.tools import load_memory, preload_memory
from google.genai import types

print("✅ ADK components imported successfully.")

✅ ADK components imported successfully.
```

+ Code + Markdown

1.4: Helper functions

This helper function manages a complete conversation session, handling session creation/retrieval, query processing, and response streaming.

Misterios

Fragmento de código

from kaggle_secrets import UserSecretsClient
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0

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Code

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Process each query

for query in user_queries:

print(f"\nUser > {query}")

query_content = types.Content(role="user", parts=[types.Part(text=query)])

Stream agent response

async for event in runner_instance.run_async(

user_id=USER_ID, session_id=session.id, new_message=query_content

):

if event.is_final_response() and event.content and event.content.parts:

text = event.content.parts[0].text

if text and text != "None":

print(f"Model: > {text}")

print(✅ Helper functions defined.)

Helper functions defined.

+ Code + Markdown

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets

user_secrets = UserSecretsClient()

secret_value_0 = user_secrets.get_secr

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Code

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retry_config = types.HttpRetryOptions(

attempts=5, # Maximum retry attempts

exp_base=7, # Delay multiplier

initial_delay=1,

http_status_codes=[429, 500, 503, 504], # Retry on these HTTP errors

)

+ Code + Markdown

Section 2: Memory Workflow

From the Introduction section, you now know why we need Memory. In order to integrate Memory into your Agents, there are **three high-level steps**.

Three-step integration process:

1. Initialize → Create a MemoryService, and provide it to your agent via the Runner

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets

user_secrets = UserSecretsClient()

secret_value_0 = user_secrets.get_secr

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Next, create a simple agent to answer user queries.

```
# Define constants used throughout the notebook
APP_NAME = "MemoryDemoApp"
USER_ID = "demo_user"

# Create agent
user_agent = LlmAgent(
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    name="MemoryDemoAgent",
    instruction="Answer user questions in simple words.",
)

print("✅ Agent created")
```

✅ Agent created

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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```
# Create Session Service
session_service = InMemorySessionService() # Handles conversations

# Create runner with BOTH services
runner = Runner(
    agent=user_agent,
    app_name="MemoryDemoApp",
    session_service=session_service,
    memory_service=memory_service, # Memory service is now available!
)

print("✅ Agent and Runner created with memory support!")
```

✅ Agent and Runner created with memory support!

+ Code + Markdown

Important

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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it available for future searches.

```
# This is the key method!
await memory_service.add_session_to_memory(session)

print("✅ Session added to memory!")
```

✅ Session added to memory!

+ Code+ Markdown

Section 5: Enable Memory Retrieval in Your Agent

You've successfully transferred session data to memory, but there's one crucial step remaining. **Agents can't directly access the MemoryService - they need tools to search it.**

This is by design: it gives you control over when and how memory is retrieved.

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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```
# Create agent
user_agent = LlmAgent(
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    name="MemoryDemoAgent",
    instruction="Answer user questions in simple words. Use load_memory tool if you need to search memory.",
    tools=[
        load_memory
    ], # Agent now has access to Memory and can search it whenever it decides to!
)

print("✅ Agent with load_memory tool created.")
```

✅ Agent with load_memory tool created.

+ Code+ Markdown

5.3 Update the Runner and Test

Let's now update the Runner to use our new `user_agent` that has the `load_memory` tool. And we'll ask the Agent

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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```
# Create a new runner with the updated agent
runner = Runner(
    agent=user_agent,
    app_name=APP_NAME,
    session_service=session_service,
    memory_service=memory_service,
)

await run_session(runner, "What is my favorite color?", "color-test")

### Session: color-test

User > What is my favorite color?

ClientError                                Traceback (most recent call last)
/tmp/ipykernel_49/3442372156.py in <cell line: 1>()
      7 )
      8
----> 9 await run_session(runner, "What is my favorite color?", "color-test")
```

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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```
await run_session(runner, "My birthday is on March 15th.", "birthday-session-01")

### Session: birthday-session-01

User > My birthday is on March 15th.
Model: > Okay, I will remember that your birthday is on March 15th.
```

+ Code + Markdown

Now manually save this session to memory. This is the crucial step that transfers the conversation from short-term session storage to long-term memory storage.

+ Code + Markdown

[]:

```
# Manually save the session to memory
birthday_session = await session_service.get_session(
    app_name=APP_NAME, user_id=USER_ID, session_id="birthday-session-01"
)

await memory_service.add_session_to_memory(birthday_session)
```

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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Day 3b - Agent Memory

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```
# Manually save the session to memory
birthday_session = await session_service.get_session(
    app_name=APP_NAME, user_id=USER_ID, session_id="birthday-session-01"
)

await memory_service.add_session_to_memory(birthday_session)

print("✅ Birthday session saved to memory!")
```

+ Code+ Markdown

Here's the crucial test: we'll start a completely new session with a different session ID and ask the agent to recall the birthday.

[]:

```
# Test retrieval in a NEW session
await run_session(
    runner, "When is my birthday?", "birthday-session-02" # Different session ID
```

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr
```

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```
# Test retrieval in a NEW session
await run_session(
    runner, "When is my birthday?", "birthday-session-02" # Different session ID
)

### Session: birthday-session-02

User > When is my birthday?

ClientError                                Traceback (most recent call last)
/tmp/ipykernel_49/3619762516.py in <cell line: 1>()
      1 # Test retrieval in a NEW session
----> 2 await run_session(
      3     runner, "When is my birthday?", "birthday-session-02" # Different session ID
      4 )

/tmp/ipykernel_49/1144951127.py in run_session(runner_instance, user_queries, session_id)
     25
     26     # Stream agent response
--> 27     async for event in runner_instance.run_async(
     28         user_id=USER_ID, session_id=session.id, new_message=query_content
     29     ):
```

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr
```

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```
# Search for color preferences
search_response = await memory_service.search_memory(
    app_name=APP_NAME, user_id=USER_ID, query="What is the user's favorite color?"
)

print("🔍 Search Results:")
print(f" Found {len(search_response.memories)} relevant memories")
print()

for memory in search_response.memories:
    if memory.content and memory.content.parts:
        text = memory.content.parts[0].text[:80]
        print(f" [{memory.author}]: {text}...")
```

🔍 Search Results:
Found 4 relevant memories

[user]: My favorite color is blue-green. Can you write a Haiku about it?...

[MemoryDemoAgent]: A calming deep shade,
Like the ocean and the trees,
My favorite color....

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0

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```
async def auto_save_to_memory(callback_context):
    """Automatically save session to memory after each agent turn."""
    await callback_context._invocation_context.memory_service.add_session_to_memory(
        callback_context._invocation_context.session
    )

print("✅ Callback created.")
```

✅ Callback created.

6.3 Create an Agent: Callback and PreLoad Memory Tool

Now create an agent that combines:

- Automatic storage: `after_agent_callback` saves conversations
- Automatic retrieval: `preload_memory` loads memories

This creates a fully automated memory system with zero manual intervention.

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0

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Automatic retrieval: preload_memory loads memories

This creates a fully automated memory system with zero manual intervention.

```
# Agent with automatic memory saving
auto_memory_agent = LlmAgent(
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    name="AutoMemoryAgent",
    instruction="Answer user questions.",
    tools=[preload_memory],
    after_agent_callback=auto_save_to_memory, # Saves after each turn!
)

print("✅ Agent created with automatic memory saving!")
```

✅ Agent created with automatic memory saving!

+ Code + Markdown

What happens automatically:

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0

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0.4 Create a runner and test the Agent

Time to test! Create a Runner with the auto-memory agent, connecting the session and memory services.

```
# Create a runner for the auto-save agent
# This connects our automated agent to the session and memory services
auto_runner = Runner(
    agent=auto_memory_agent, # Use the agent with callback + preload_memory
    app_name=APP_NAME,
    session_service=session_service, # Same services from Section 3
    memory_service=memory_service,
)

print("✅ Runner created.")
```

✅ Runner created.

+ Code + Markdown

[]: # Test 1: Tell the agent about a gift (first conversation)
The callback will automatically save this to memory when the turn completes

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0

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Congratulations! You've learned Memory Management in ADK!

📖 Learn More:

- ADK Memory Documentation
- Vertex AI Memory Bank
- Memory Consolidation Guide

🔗 Next Steps:

Ready for Day 4? Learn how to **implement Observability and Evaluate your agents** to ensure they're working as intended in production!

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecretsClient
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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Registro

que su agente recuerde el contexto y tenga conversaciones coherentes de múltiples turnos. En el segundo cuaderno, le dará a su agente memoria a largo plazo que persista en diferentes sesiones.

Tarea del Día 3:

Complete la Unidad 3 - "Ingeniería de Contexto: Sesiones y Memoria".

- ☒ Escucha el episodio del podcast que resume esta unidad.
- ☒ Para complementar el podcast, lea el documento técnico "Ingeniería de contexto: sesiones y memoria".
- ☒ Complete estos codelabs en Kaggle:
 - ☒ Construye agentes con estado y realiza ingeniería de contexto.
 - ☒ Explora cómo usar la memoria con tu agente.

[Opcional]: Vea la transmisión en vivo grabada de YouTube. Kanchana Patilola y Anant Nawalgaria estuvieron acompañados por el autor del codelab, Kristopher Overholt, junto con invitados de Google: Steven Johnson, Kimberly Milam y Julia Wiesinger, y el orador externo Jay Alamar de Cohere.

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Whitepaper Companion Podcast

Agent Quality

A Practical Guide from Evaluation to Observability

Authors: Meltem Subasioglu, Turan Bulmus, and Wafae Bakkali

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Calidad del agente

Autores: Meltem Subasioglu, Turan Bulmus y Wafae Bakkali

Introducción

Esta es una guía práctica que abarca desde la evaluación hasta la observabilidad. Nos encontramos en los albores de la era de los agentes. La transición de herramientas predecibles basadas en instrucciones a agentes de IA autónomos y orientados a objetivos representa uno de los cambios más profundos en la ingeniería de software en décadas. Si bien estos agentes desbloquean capacidades increíbles, su indeterminismo inherente los hace impredecibles y desbarata nuestros modelos tradicionales de garantía de calidad.

Este documento técnico sirve como guía práctica para esta nueva realidad, basada en un principio simple pero radical: la calidad del agente es un pilar arquitectónico, no una fase final de pruebas.

Lea el informe técnico a continuación.

DIA 4 A

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+ Code + Markdown

```
import os
from kaggle_secrets import UserSecretsClient

try:
    GOOGLE_API_KEY = UserSecretsClient().get_secret("GOOGLE_API_KEY")
    os.environ["GOOGLE_API_KEY"] = GOOGLE_API_KEY
    print("✅ Setup and authentication complete.")
except Exception as e:
    print(f"🚨 Authentication Error: Please make sure you have added 'GOOGLE_API_KEY' to yo
    )

✅ Setup and authentication complete.
```

+ Code + Markdown

1.3: Set up logging and cleanup old files

Let's configure logging for our debugging session. The following cell makes sure we also capture other log levels, like

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr
```

Copiar al portapapeles

GOOGLE_API_KEY *****

+ Agregar secreto Copiar fragmento

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```
import logging
import os

# Clean up any previous logs
for log_file in ["logger.log", "web.log", "tunnel.log"]:
    if os.path.exists(log_file):
        os.remove(log_file)
        print(f"🧹 Cleaned up {log_file}")

# Configure logging with DEBUG log level.
logging.basicConfig(
    filename="logger.log",
    level=logging.DEBUG,
    format="%(filename)s:%(lineno)s %(levelname)s: %(message)s",
)

print("✅ Logging configured")

✅ Logging configured
```

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr
```

Copiar al portapapeles

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```
from IPython.core.display import display, HTML
from jupyter_server.serverapp import list_running_servers

# Gets the proxied URL in the Kaggle Notebooks environment
def get_adk_proxy_url():
    PROXY_HOST = "https://kkg-production.jupyter-proxy.kaggle.net"
    ADK_PORT = "8000"

    servers = list(list_running_servers())
    if not servers:
        raise Exception("No running Jupyter servers found.")

    baseUrl = servers[0]["baseUrl"]

    try:
        path_parts = baseUrl.split("/")
        kernel = path_parts[2]
        token = path_parts[3]
    except IndexError:
        raise Exception(f"Could not parse kernel/token from base URL: {baseUrl}")
```

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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```
<em style="font-size: 0.9em; color: #555;">(If you click the button before run
</div>
<a href='url' target='_blank' style="
    display: inline-block; background-color: #1a73e8; color: white; padding: 10px;
    text-decoration: none; border-radius: 25px; font-family: sans-serif; font-weight: bold;
    box-shadow: 0 2px 5px rgba(0,0,0,0.2); transition: all 0.2s ease;">
    Open ADK Web UI (after running cell below) </a>
</div>

display(HTML(styled_html))

return url_prefix

print('✅ Helper functions defined.')
```

✅ Helper functions defined.

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

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+ Code

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```
!sdk create research-agent --model gemini-2.5-flash-lite --api_key $GOOGLE_API_KEY
```

Agent created in /kaggle/working/research-agent:

- .env
- __init__.py
- agent.py

Agent definition

Next, let's create our root agent.

- We'll configure it as an `LlmAgent`, give it a name, model and instruction.
- The `root_agent` gets the user prompt and delegates the search to the `google_search_agent`.
- Then, the agent uses the `count_papers` tool to count the number of papers returned.

Pay attention to the root agent's instructions and the `count_papers` tool parameter!

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

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```
tools=[google_search]

)

# Root agent
root_agent = LlmAgent(
    name="research_paper_finder_agent",
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    instruction="Your task is to find research papers and count them.

    You MUST ALWAYS follow these steps:
    1) Find research papers on the user provided topic using the 'google_search_agent'.
    2) Then, pass the papers to 'count_papers' tool to count the number of papers returned.
    3) Return both the list of research papers and the total number of papers.
    """
    tools=[AgentTool(agent=google_search_agent), count_papers]
)

Overwriting research-agent/agent.py
```

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

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url_prefix = get_adk_proxy_url()

IMPORTANT: Action Required

The ADK web UI is **not running yet**. You must start it in the next cell.

1. Run the next cell (the one with `!adk web ...`) to start the ADK web UI.

2. Wait for that cell to show it is "Running" (it will not "complete").

3. Once it's running, **return to this button** and click it to open the UI.

(If you click the button before running the next cell, you will get a 500 error.)

Open ADK Web UI (after running cell below)

+ Code

+ Markdown

Now you can start the ADK web UI with the `--log_level` parameter.

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Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

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[7]:

!adk web --log_level DEBUG --url_prefix {url_prefix}

```
/usr/local/lib/python3.11/dist-packages/google/adk/c11/fast_api.py:130: UserWarning: [EXPERIMENTAL] InMemoryCredentialService: This feature is experimental and may change or be removed in future versions without notice. It may introduce breaking changes at any time.
credential_service = InMemoryCredentialService()
/usr/local/lib/python3.11/dist-packages/google/adk/auth/credential_service/in_memory_credential_s
ervice.py:33: UserWarning: [EXPERIMENTAL] BaseCredentialService: This feature is experimental and
may change or be removed in future versions without notice. It may introduce breaking changes at
any time.
super().__init__()
INFO: Started server process [90]
INFO: Waiting for application startup.

ADK Web Server started
For local testing, access at http://127.0.0.1:8080.
```

+ Code

+ Markdown

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Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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GOOGLE_API_KEY

+ Agregar secreto

Copiar fragmento

Agent Development Kit

research-agent

Trace Events State Artifacts Sessions Eval

No invocations found

SESSION ID eb4b3f5d-26bf-4be0-b46c-a30715b705c7 USER ID user:user Token Streaming New Session

Type a Message...

Day 4a - Agent Observability

Borrador guardado

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Run All Code Sesión de borrador Cargando 0 m

```
# Callback 1: Runs before an agent is called. You can add any custom logic here.
async def before_agent_callback(
    self, *, agent: BaseAgent, callback_context: CallbackContext
) -> None:
    """Count agent runs."""
    self.agent_count += 1
    logging.info(f"[Plugin] Agent run count: {self.agent_count}")

# Callback 2: Runs before a model is called. You can add any custom logic here.
async def before_model_callback(
    self, *, callback_context: CallbackContext, llm_request: LlmRequest
) -> None:
    """Count LLM requests."""
    self.llm_request_count += 1
    logging.info(f"[Plugin] LLM request count: {self.llm_request_count}")

----- EXAMPLE PLUGIN - DOES NOTHING -----
```

+ Code + Markdown

Key insight: You register a plugin **once** on your runner, and it automatically applies to **every agent, tool call, and LLM request** in your system as per your definition. Read more about Plugin hooks [here](#).

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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Day 4a - Agent Observability

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```
description= searches for information using google search ,
instruction="Use the google_search tool to find information on the given topic. Return
tools=[google_search],
)

# Root agent
research_agent_with_plugin = LlmAgent(
    name="research_paper_finder_agent",
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    instruction="Your task is to find research papers and count them.

You must follow these steps:
1) Find research papers on the user provided topic using the 'google_search_agent'.
2) Then, pass the papers to 'count_papers' tool to count the number of papers returned
3) Return both the list of research papers and the total number of papers.
"""
    tools=[AgentTool(agent=google_search_agent), count_papers],
)

print('✅ Agent created')
```

✅ Agent created

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr
```

Copiar al portapapeles

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Day 4a - Agent Observability

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```
from google.adk.runners import InMemoryRunner
from google.adk.plugins.logging_plugin import (
    LoggingPlugin,
) # <---- 1. Import the Plugin
from google.genai import types
import asyncio

runner = InMemoryRunner(
    agent=research_agent_with_plugin,
    plugins=[
        LoggingPlugin()
    ], # <---- 2. Add the plugin. Handles standard Observability logging across ALL agents
)

print('✅ Runner configured')
```

✅ Runner configured

+ Code + Markdown

Let's now run the agent using `run_debug` function.

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Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr
```

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print("🚀 Running agent with LoggingPlugin...")
print("📄 Watch the comprehensive logging output below:\n")

response = await runner.run_debug("Find recent papers on quantum computing")

Running agent with LoggingPlugin...
📄 Watch the comprehensive logging output below:

Created new session: debug_session_id

User > Find recent papers on quantum computing
[logging_plugin] 🚀 USER MESSAGE RECEIVED
[logging_plugin] Invocation ID: e-38b2a5cd-b617-4172-ae5d-9a856f14228c
[logging_plugin] Session ID: debug_session_id
[logging_plugin] User ID: debug_user_id
[logging_plugin] App Name: InMemoryRunner
[logging_plugin] Root Agent: research_paper_finder_agent
[logging_plugin] User Content: text: "Find recent papers on quantum computing"
[logging_plugin] 🚀 INVOCATION STARTING
[logging_plugin] Invocation ID: e-38b2a5cd-b617-4172-ae5d-9a856f14228c

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0

Copiar al portapapeles

GOOGLE_API_KEY

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 Congratulations!

You now know how to:

- ✅ Debug agent failures through DEBUG logs and the ADK web UI
- ✅ Use the core debugging pattern: symptom → logs → root cause → fix
- ✅ Scale observability with `LoggingPlugin` for production systems
- ✅ Understand when to use the different logging types

Note: No submission required!

This notebook is for your hands-on practice and learning only. You **do not** need to submit it anywhere to complete the course.

+ Code + Markdown

Resources

Refer to the ADK documentation to learn more about observability:

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0

Copiar al portapapeles

GOOGLE_API_KEY

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4 DIA B

Curso intensivo de 5

Día 4b - Evaluación

Kaggle Notebook Ed

Día 4a - Observabili

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+ Code

+ Markdown

```
import os
from kaggle_secrets import UserSecretsClient

try:
    GOOGLE_API_KEY = UserSecretsClient().get_secret("GOOGLE_API_KEY")
    os.environ["GOOGLE_API_KEY"] = GOOGLE_API_KEY
    print("✅ Setup and authentication complete.")
except Exception as e:
    print(f"❌ Authentication Error: Please make sure you have added 'GOOGLE_API_KEY' to yo
)

✅ Setup and authentication complete.
```

1.3: Set up proxy and tunneling

We'll use a proxy to access the ADK web UI from within the Kaggle Notebooks environment. If you are running this outside the Kaggle environment, you don't need to do this.

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr

Copiar al portapapeles

GOOGLE_API_KEY

+ Agregar secreto

Copiar fragmento

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+ Code

+ Markdown

```
</ol>
<em style="font-size: 0.9em; color: #555;">(If you click the button before ru
</div>
<a href='url' target='_blank' style="
display: inline-block; background-color: #1a73e8; color: white; padding: 10px
text-decoration: none; border-radius: 25px; font-family: sans-serif; font-we
box-shadow: 0 2px 5px rgba(0,0,0,0.2); transition: all 0.2s ease;">
Open ADK Web UI (after running cell below) </
</a>
</div>
<div>
display(HTML(styled_html))

return url_prefix

print("✅ Helper functions defined.")

✅ Helper functions defined.
```

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr

Copiar al portapapeles

GOOGLE_API_KEY

+ Agregar secreto

Copiar fragmento

Day 4b - Agent Evaluation

Always try to be helpful and control whatever device the user asks for.

When users ask about device capabilities, tell them about all the amazing features you

tools=[set_device_status],

Having trouble finding a file? You'll find some useful tips in our Finding Your Files tutorial.

Writing home_automation_agent/agent.py

```
FileNotFoundError                                Traceback (most recent call last)
/tmp/ipykernel_49/1562375946.py in <cell line: 0>()
----> 1 get_ipython().run_cell_magic('writefile', 'home_automation_agent/agent.py', '\nfrom googl
e.adk.agents import LlmAgent\nfrom google.adk.models.google_llm import Gemini\n\nfrom google.genai
import types\n\n# Configure Model Retry on errors\nretry_config = types.HttpRetryOptions(\n    att
empts=5, # Maximum retry attempts\n    exp_base=7, # Delay multiplier\n    initial_delay=1,\n    http_status_codes=[429, 500, 503, 504], # Retry on these HTTP errors\n)\n\ndef set_device_status
(location: str, device_id: str, status: str) -> dict:\n    """Sets the status of a smart home devi
ce.\n\n    Args:\n        location: The room where the device is located.\n        device_id: The
unique identifier for the device.\n        status: The desired status, either 'ON' or 'OFF'.
\n\n    Returns:\n        A dictionary confirming the action.\n    """\n    print(f"Tool Call: Setti
ng (device_id) in (location) to (status)")\n    return {\n        "success": True,\n        "messa
ge": f"Successfully set the {device_id} in {location} to {status.lower()}."
    }\n\n# This agent
```

Day 4b - Agent Evaluation

Borrador guardado

Get the proxied URL to access the ADK web UI in the Kaggle Notebooks environment:

```
[8]: url_prefix = get_adk_proxy_url()
```

IMPORTANT: Action Required

The ADK web UI is **not running yet**. You must start it in the next cell.

- Run the next cell (the one with `!adk web ...`) to start the ADK web UI.
- Wait for that cell to show it is "Running" (it will not "complete").
- Once it's running, **return to this button** and click it to open the UI.

(If you click the button before running the next cell, you will get a 500 error.)

Open ADK Web UI (after running cell below)

Day 4b - Agent Evaluation Borrador guardado

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```
!adk web --url_prefix {url_prefix}

/usr/local/lib/python3.11/dist-packages/google/adk/cli/fast_api.py:130: UserWarning: [EXPERIMENTAL] InMemoryCredentialService: This feature is experimental and may change or be removed in future versions without notice. It may introduce breaking changes at any time.
  credential_service = InMemoryCredentialService()
/usr/local/lib/python3.11/dist-packages/google/adk/auth/credential_service/in_memory_credential_service.py:33: UserWarning: [EXPERIMENTAL] BaseCredentialService: This feature is experimental and may change or be removed in future versions without notice. It may introduce breaking changes at any time.
  super().__init__()
INFO: Started server process [114]
INFO: Waiting for application startup.

ADK Web Server started
For local testing, access at http://127.0.0.1:8080.
```

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecretsClient
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

Copiar al portapapeles

GOOGLE_API_KEY *****

+ Agregar secreto Copiar fragmento

Day 4b - Agent Evaluation Borrador guardado

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+ Run All Code Sesión de borrador Cargando 0 m

```
print(✅ Evaluation configuration created!)
print("\n📋 Evaluation Criteria:")
print("• tool_trajectory_avg_score: 1.0 - Requires exact tool usage match")
print("• response_match_score: 0.8 - Requires 80% text similarity")
print("\n🔍 What this evaluation will catch:")
print(✅ Incorrect tool usage (wrong device, location, or status)")
print(✅ Poor response quality and communication")
print(✅ Deviations from expected behavior patterns")

Having trouble finding a file? You'll find some useful tips in our Finding Your Files tutorial.

FileNotFoundError Traceback (most recent call last)
/tmp/ipykernel_49/2196862736.py in <cell line: 0>()
    9 }
   10
--> 11 with open("home_automation_agent/test_config.json", "w") as f:
   12     json.dump(eval_config, f, indent=2)
   13

FileNotFoundError: [Errno 2] No such file or directory: 'home_automation_agent/test_config.json'
```

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecretsClient
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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GOOGLE_API_KEY *****

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Day 4b - Agent Evaluation

```
# Create evaluation test cases that reveal tool usage and response quality problems
test_cases = {
    "eval_set_id": "home_automation_integration_suite",
    "eval_cases": [
        {
            "eval_id": "living_room_light_on",
            "conversation": [
                {
                    "user_content": {
                        "parts": [
                            {"text": "Please turn on the floor lamp in the living room"}
                        ]
                    },
                    "final_response": {
                        "parts": [
                            {
                                "text": "Successfully set the floor lamp in the living room"
                            }
                        ]
                    },
                    "intermediate_data": {
                        "tool_uses": [

```

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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Day 4b - Agent Evaluation

```
print("\n Expected results:")
print("\n basic_device_control: Should pass both criteria")
print("\n wrong_tool_usage_test: May fail tool_trajectory if agent uses wrong parameters")
print("\n poor_response_quality_test: May fail response_match if response differs too much")
```

Having trouble finding a file? You'll find some useful tips in our [Finding Your Files tutorial](#).

```
FileNotFoundError: Traceback (most recent call last)
/tmp/ipykernel_49/1391270917.py in <cell line: 0>()
      1 import json
      2
----> 3 with open("home_automation_agent/integration.evalset.json", "w") as f:
      4     json.dump(test_cases, f, indent=2)
      5

FileNotFoundError: [Errno 2] No such file or directory: 'home_automation_agent/integration.evalset.json'
```

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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4.3: Run CLI Evaluation

Execute the `adk eval` command, pointing it to your agent directory, the evalset, and the config file.

```
print("🚀 Run this command to execute evaluation:")
!adk eval home_automation_agent home_automation_agent/integration.evalset.json --config.f:
```

🚀 Run this command to execute evaluation:
Usage: adk eval [OPTIONS] AGENT_MODULE_FILE_PATH [EVAL_SET_FILE_PATH_OR_ID]...
Try 'adk eval --help' for help.

Error: Invalid value for 'AGENT_MODULE_FILE_PATH': Directory 'home_automation_agent' does not exist.

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4.4: Analyzing sample evaluation results

The command will run all test cases and print a summary. The `--print_detailed_results` flag provides a turn-by-

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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```
print("✅ tool_trajectory_avg_score: 1.0/1.0")
print()
print("📋 What this tells us:")
print("• TOOL USAGE: Perfect - Agent used correct tool with correct parameters")
print("• RESPONSE QUALITY: Poor - Response text too different from expected")
print("• ROOT CAUSE: Agent's communication style, not functionality")
print()
print("🔍 ACTIONABLE INSIGHTS:")
print("1. Technical capability works (tool usage perfect)")
print("2. Communication needs improvement (response quality failed)")
print("3. Fix: Update agent instructions for clearer language or constrained response.")
print()
```

📊 Understanding Evaluation Results:

🔍 EXAMPLE ANALYSIS:

Test Case: living_room_light_on

- ❌ response_match_score: 0.45/0.80
- ✅ tool_trajectory_avg_score: 1.0/1.0

📋 What this tells us:

- TOOL USAGE: Perfect - Agent used correct tool with correct parameters

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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Congratulations!

You've learned

- Interactive test creation and analysis in the ADK web UI
- Tool trajectory and response metrics
- Automated regression testing using `adk eval` CLI command
- How to analyze evaluation results and fix agents based on it

Note: No submission required!

This notebook is for your hands-on practice and learning only. You **do not** need to submit it anywhere to complete the course.

Resources

- [ADK Evaluation overview](#)
- [Different evaluation criteria](#)

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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Tarea del Día 4:

Complete la Unidad 4 - "Calidad del Agente".

- Escucha el episodio del podcast que resume esta unidad.
- Para complementar el podcast, lea el [informe técnico sobre la calidad del agente](#).
- Completa estos codelabs en Kaggle:
 - Implementa la observabilidad para ayudarte a depurar tus agentes.
 - Evalúe a sus agentes.

[Opcional]: Vea la [transmisión en vivo grabada de YouTube](#). Kanchana Patlolla y Anant Nawalgaria estuvieron acompañados por la autora del codelab, Sita Lakshmi Sangameswaran, junto con invitados de Google: Wafae Bakkali, Turan Bulmus y Sian Gooding, y el invitado externo Jiwei Liu de NVIDIA.

DIA 5

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Del prototipo a la producción

Autores: Sokratis Kartakis, Gabriela Hernández Larios, Ran Li, Elía Secchi y Huang Xia

Introducción

Este documento técnico ofrece una guía completa sobre el ciclo de vida operativo de los agentes de IA, centrándose en el despliegue, la escalabilidad y la puesta en producción. Partiendo de la información del cuarto día sobre evaluación y observabilidad, esta guía destaca cómo generar la confianza necesaria para implementar agentes en producción mediante sólidas canalizaciones de CI/CD e infraestructura escalable. Explora los desafíos de la transición de sistemas basados en agentes desde prototipos a soluciones empresariales, con especial atención a la interoperabilidad entre agentes (A2A). Esta guía ofrece información práctica para ingenieros de IA/ML, profesionales de DevOps y arquitectos de sistemas.

Lea el informe técnico a continuación.

DIA 5 A

The image displays two screenshots of a Kaggle Notebook Editor interface, showing the progression of code execution for a project titled "Day 5a - Agent2Agent Comunicación".

Top Screenshot:

- Code Cell:** The code imports `os` and `UserSecretsClient` from `kaggle_secrets`. It attempts to retrieve the `GOOGLE_API_KEY` secret and set the environment variable. A green checkmark indicates "Setup and authentication complete.".
- Output:** A green checkmark and the text "Setup and authentication complete." are displayed.
- Right Panel:** The "Misterios" (Secrets) panel shows the `GOOGLE_API_KEY` secret, which has been successfully added and is masked with asterisks.

Bottom Screenshot:

- Code Cell:** The code imports various components from the Agent Development Kit (ADK) and the Generative AI library, including `LlmAgent`, `RemoteA2aAgent`, `to_a2a`, `Gemini`, `Runner`, `InMemorySessionService`, and `types`. It also imports `warnings` and filters them to ignore. A green checkmark indicates "ADK components imported successfully.".
- Output:** A green checkmark and the text "ADK components imported successfully." are displayed.
- Right Panel:** The "Misterios" (Secrets) panel shows the `GOOGLE_API_KEY` secret, which has been successfully added and is masked with asterisks.

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Code

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↑

↓

🗑

retry_config = types.HttpRetryOptions(
 attempts=5, # Maximum retry attempts
 exp_base=7, # Delay multiplier
 initial_delay=1,
 http_status_codes=[429, 500, 503, 504], # Retry on these HTTP errors
)

+ Code+ Markdown

📖 Section 1: Create the Product Catalog Agent (To Be Exposed)

We'll create a **Product Catalog Agent** that provides product information from an external vendor's catalog. This agent will be **exposed via A2A** so other agents (like customer support) can use it.

Why expose this agent?
• In a real system, this would be maintained by an **external vendor** or third-party provider

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr

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↑

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🗑

from IPython.display import Image, HTML
description="External vendor's product catalog agent that provides product information"
instruction="""
You are a product catalog specialist from an external vendor.
When asked about products, use the get_product_info tool to fetch data from the catalog.
Provide clear, accurate product information including price, availability, and specs.
If asked about multiple products, look up each one.
Be professional and helpful.
""",
tools=[get_product_info], # Register the product lookup tool
)

print("✅ Product Catalog Agent created successfully!")
print(" Model: gemini-2.5-flash-lite")
print(" Tool: get_product_info()")
print(" Ready to be exposed via A2A...")

+ Code+ Markdown

✅ Product Catalog Agent created successfully!
Model: gemini-2.5-flash-lite
Tool: get_product_info()
Ready to be exposed via A2A...

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Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr

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```
print(f" Agent card: http://localhost:8001/.well-known/agent-card.json")
break
except requests.exceptions.RequestException:
    time.sleep(5)
    print(".", end="", flush=True)
else:
    print("\n⚠️ Server may not be ready yet. Check manually if needed.")

# Store the process so we can stop it later
globals()["product_catalog_server_process"] = server_process

Product Catalog agent code saved to /tmp/product_catalog_server.py
Starting Product Catalog Agent server...
Waiting for server to be ready...
...
Product Catalog Agent server is running!
Server URL: http://localhost:8001
Agent card: http://localhost:8001/.well-known/agent-card.json
```

+ Code + Markdown

View the Auto-Generated Agent Card

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr
```

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```
# Fetch the agent card from the running server
try:
    response = requests.get(
        "http://localhost:8001/.well-known/agent-card.json", timeout=5
    )

    if response.status_code == 200:
        agent_card = response.json()
        print("📄 Product Catalog Agent Card:")
        print(json.dumps(agent_card, indent=2))

        print("\n🔑 Key Information:")
        print(f"   Name: {agent_card.get('name')}")
        print(f"  Description: {agent_card.get('description')}")
        print(f"    URL: {agent_card.get('url')}")
        print(f" Skills: {len(agent_card.get('skills', []))} capabilities exposed")
    else:
        print(f"❌ Failed to fetch agent card: {response.status_code}")

except requests.exceptions.RequestException as e:
    print(f"❌ Error fetching agent card: {e}")
    print("Make sure the Product Catalog Agent server is running (previous cell)")
```

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr
```

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```
Product Catalog Agent Card:
{
  "capabilities": {},
  "defaultInputModes": [
    "text/plain"
  ],
  "defaultOutputModes": [
    "text/plain"
  ],
  "description": "External vendor's product catalog agent that provides product information and availability.",
  "name": "product_catalog_agent",
  "preferredTransport": "JSONRPC",
  "protocolVersion": "0.3.0",
  "skills": [
    {
      "description": "External vendor's product catalog agent that provides product information and availability. \n I am a product catalog specialist from an external vendor.\n When asked about products, use the get_product_info tool to fetch data from the catalog.\n Provide clear, accurate product information including price, availability, and specs.\n If asked about multiple products, look up each one.\n Be professional and helpful.\n ",
      "id": "product_catalog_agent",
      "name": "model",
      "tags": [
        "llm"
      ]
    }
  ]
}
```

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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```
# Create a RemoteA2aAgent that connects to our Product Catalog Agent
# This acts as a client-side proxy - the Customer Support Agent can use it like a local agent
remote_product_catalog_agent = RemoteA2aAgent(
    name="product_catalog_agent",
    description="Remote product catalog agent from external vendor that provides product information and availability.",
    # Point to the agent card URL - this is where the A2A protocol metadata lives
    agent_card=f"http://localhost:8001{AGENT_CARD_WELL_KNOWN_PATH}",
)

print("Remote Product Catalog Agent proxy created!")
print(f"Connected to: http://localhost:8001")
print(f"Agent card: http://localhost:8001{AGENT_CARD_WELL_KNOWN_PATH}")
print("The Customer Support Agent can now use this like a local sub-agent!")

Remote Product Catalog Agent proxy created!
Connected to: http://localhost:8001
Agent card: http://localhost:8001/.well-known/agent-card.json
The Customer Support Agent can now use this like a local sub-agent!
```

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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```
When customers ask about products:
1. Use the product_catalog_agent sub-agent to look up product information
2. Provide clear answers about pricing, availability, and specifications
3. If a product is out of stock, mention the expected availability
4. Be helpful and professional!

Always get product information from the product_catalog_agent before answering customer questions.

sub_agents=[remote_product_catalog_agent], # Add the remote agent as a sub-agent!

print("✅ Customer Support Agent created!")
print("Model: gemini-2.5-flash-lite")
print("Sub-agents: 1 (remote Product Catalog Agent via A2A)")
print("Ready to help customers!")
```

✅ Customer Support Agent created!
Model: gemini-2.5-flash-lite
Sub-agents: 1 (remote Product Catalog Agent via A2A)
Ready to help customers!

+ Code + Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecretsClient
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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Testing A2A Communication...

Customer: Can you tell me about the iPhone 15 Pro? Is it in stock?

Support Agent response:

```
INFO:google_adk.google.adk.models.google_llm:Sending out request, model: gemini-2.5-flash-lite, ba
ckend: GoogleLLMVariant.GEMINI_API, stream: False
INFO:google_adk.google.adk.models.google_llm:Response received from the model.
WARNING:google_genai.types.Warning: there are non-text parts in the response: ['function_call'], r
eturning concatenated text result from text parts. Check the full candidates.content.parts accesso
r to get the full model response.
INFO:google_adk.google.adk.agents.remote_a2a_agent:Successfully resolved remote A2A agent: product
catalog_agent
The iPhone 15 Pro is currently in stock, with 8 units available. It is priced at $999 and features
a 128GB storage capacity with a titanium finish.
```

+ Code + Markdown

Try More Examples

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Fragmento de código

```
from kaggle_secrets import UserSecretsClient
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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```
# Test comparing multiple products
await test_a2a_communication(
    "I'm looking for a laptop. Can you compare the Dell XPS 15 and MacBook Pro 14 for me?"
)

INFO:google_adk.google.adk.models.google_llm:Sending out request, model: gemini-2.5-flash-lite, ba
ckend: GoogleLLMVariant.GEMINI_API, stream: False

Customer: I'm looking for a laptop. Can you compare the Dell XPS 15 and MacBook Pro 14 for me?

Support Agent response:
-----
INFO:google_adk.google.adk.models.google_llm:Response received from the model.
WARNING:google_genai.types.Warning: there are non-text parts in the response: ['function_call'], r
eturning concatenated text result from text parts. Check the full candidates.content.parts accesso
r to get the full model response.
I can help you with that! Here's a comparison of the Dell XPS 15 and the MacBook Pro 14:

**Dell XPS 15:**
* **Price:** $1,299
* **Availability:** In Stock (45 units available)
```

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Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr
```

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```
# Test specific product inquiry
await test_a2a_communication(
    "Do you have the Sony WH-1000XM5 headphones? What's the price?"
)

INFO:google_adk.google.adk.models.google_llm:Sending out request, model: gemini-2.5-flash-lite, ba
ckend: GoogleLLMVariant.GEMINI_API, stream: False

Customer: Do you have the Sony WH-1000XM5 headphones? What's the price?

Support Agent response:
-----
INFO:google_adk.google.adk.models.google_llm:Response received from the model.
WARNING:google_genai.types.Warning: there are non-text parts in the response: ['function_call'], r
eturning concatenated text result from text parts. Check the full candidates.content.parts accesso
r to get the full model response.
The Sony WH-1000XM5 headphones are available for $399 and are currently in stock, with 67 units av
ailable. They feature noise-canceling technology and a 30-hour battery life.
```

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr
```

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GOOGLE_API_KEY

+ Agregar secreto Copiar fragmento

DIA 5 B

The image displays two sequential screenshots of a Kaggle notebook interface, titled "Day 5b - Agent Deployment".

Top Screenshot: The notebook shows a code cell with the following imports: `import os, random, time, vertexai` and `from kaggle_secrets import UserSecretsClient`, followed by `from vertexai import agent_engines`. The output indicates "Imports completed successfully". Below the code, a section titled "1.3: Add Cloud Credentials to Secrets" provides instructions: 1. Select "Add-ons" then "Google Cloud SDK" in the top menu. 2. Click "Link Account". 3. Select a Google Cloud Account. 4. Attach to the notebook. The right sidebar shows a "Misterios" panel with a code snippet for `from kaggle_secrets import UserSecretsClient` and a "GOOGLE_API_KEY" secret.

Bottom Screenshot: The notebook shows a code cell attempting to set up cloud credentials: `# Set up Cloud Credentials in Kaggle`, `user_secrets = UserSecretsClient()`, `user_credential = user_secrets.get_gcloud_credential()`, `user_secrets.set_tensorflow_credential(user_credential)`, and `print('Cloud credentials configured')`. The output shows a "BackendError" with a traceback indicating "Google Cloud SDK credential not found. Please ensure you have a authorized Google Cloud SDK for this Notebook (in Menu, Add-ons > Google Cloud SDK)". The right sidebar remains the same as in the top screenshot.

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+Run AllCodeSesión de apagado (ejecuta una celda 0 borrador para comenzar)

```
## Set your PROJECT_ID
PROJECT_ID = "your-project-id" # TODO: Replace with your project ID
os.environ["GOOGLE_CLOUD_PROJECT"] = PROJECT_ID

if PROJECT_ID == "your-project-id" or not PROJECT_ID:
    raise ValueError("⚠ Please replace 'your-project-id' with your actual Google Cloud P

print(f"✅ Project ID set to: {PROJECT_ID}")
```

```
ValueError                                Traceback (most recent call last)
/tmp/ipykernel_49/3421347792.py in <cell line: 0>()
      4
      5 if PROJECT_ID == "your-project-id" or not PROJECT_ID:
----> 6     raise ValueError("⚠ Please replace 'your-project-id' with your actual Google Cloud Pr
      7 object ID.")
      8 print(f"✅ Project ID set to: {PROJECT_ID}")

ValueError: ⚠ Please replace 'your-project-id' with your actual Google Cloud Project ID.
```

+ Code+ Markdown

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr
```

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+Run AllCodeSesión de apagado (ejecuta una celda 0 borrador para comenzar)

```
## Create simple agent - all code for the agent will live in this directory
!mkdir -p sample_agent

print(f"✅ Sample Agent directory created")
```

```
✅ Sample Agent directory created
```

+ Code+ Markdown

2.2: Create requirements file

The Agent Engine builds a dedicated environment for your agent. To ensure it runs correctly, we must declare our dependencies.

We will write a `requirements.txt` file containing the Python packages needed for the agent.

[]: %writefile sample_agent/requirements.txt

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr
```

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Day 5b - Agent Deployment

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Compartir Guardar versión 0

+ dependencies.

We will write a `requirements.txt` file containing the Python packages needed for the agent.

%writefile sample_agent/requirements.txt

google-adk
opentelemetry-instrumentation-google-genai

Overwriting sample_agent/requirements.txt

+ Code + Markdown

2.3: Create environment configuration

We need to provide the agent with the necessary cloud configuration settings.

We will write a `.env` file that sets the cloud location to `global` and explicitly enables the Vertex AI backend for the ADK SDK.

Sesión de apagado (ejecuta una celda 0 borrador para comenzar)

Run All

Code

↑ ↓

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0

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ADK SDK.

%writefile sample_agent/.env

https://cloud.google.com/vertex-ai/generative-ai/docs/learn/locations#global-endpoint
GOOGLE_CLOUD_LOCATION="global"

Set to 1 to use Vertex AI, or 0 to use Google AI Studio
GOOGLE_GENAI_USE_VERTEXAI=1

Writing sample_agent/.env

+ Code + Markdown

Configuration explained:

- GOOGLE_CLOUD_LOCATION="global" - Uses the `global` endpoint for Gemini API calls
- GOOGLE_GENAI_USE_VERTEXAI=1 - Configures ADK to use Vertex AI instead of Google AI Studio

Sesión de apagado (ejecuta una celda 0 borrador para comenzar)

Run All

Code

↑ ↓

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0

Copiar al portapapeles

GOOGLE_API_KEY *****

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Day 5b - Agent Deployment Borrador guardado

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Run All Code Sesión de borrador apagado (ejecuta una celda o borrador para comenzar)

```
status : error ,
"error_message": f"Weather information for '{city}' is not available. Try: {a

}

root_agent = Agent(
    name="weather_assistant",
    model="gemini-2.5-flash-lite", # Fast, cost-effective Gemini model
    description="A helpful weather assistant that provides weather information for cities
    instruction="""
    You are a friendly weather assistant. When users ask about the weather:

    1. Identify the city name from their question
    2. Use the get_weather tool to fetch current weather information
    3. Respond in a friendly, conversational tone
    4. If the city isn't available, suggest one of the available cities

    Be helpful and concise in your responses.
    """
    tools=[get_weather]
)
```

Writing sample_agent/agent.py

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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GOOGLE_API_KEY *****

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Day 5b - Agent Deployment Borrador guardado

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Run All Code Sesión de borrador apagado (ejecuta una celda o borrador para comenzar)

5.1: Create deployment configuration

The .agent_engine_config.json file controls the deployment settings.

```
%%writefile sample_agent/.agent_engine_config.json
{
    "min_instances": 0,
    "max_instances": 1,
    "resource_limits": {"cpu": "1", "memory": "1Gi"}
}
```

Writing sample_agent/.agent_engine_config.json

+ Code + Markdown

Configuration explained:

- "min_instances": 0 - Scales down to zero when not in use (saves costs)
- "max_instances": 1 - Maximum of 1 instance running (sufficient for this demo)
- "cpu": "1" - 1 CPU core per instance
- "memory": "1Gi" - 1 GB of memory per instance

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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These settings keep costs minimal while providing adequate resources for our weather agent.

3.2: Select deployment region

Agent Engine is available in specific regions. We'll randomly select one for this demo.

```
regions_list = ["europe-west1", "europe-west4", "us-east4", "us-west1"]
deployed_region = random.choice(regions_list)

print(f"Selected deployment region: {deployed_region}")
```

Selected deployment region: us-east4

+ Code + Markdown

About regions:

Agent Engine is available in multiple regions. For production:

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Compartir Guardar versión 0

Staging all files in: /kaggle/working/sample_agent_tmp20260505_074456
Copying agent source code...
Copying agent source code complete.
Resolving files and dependencies...
Reading agent engine config from sample_agent/.agent_engine_config.json
Reading environment variables from /kaggle/working/sample_agent/.env
Ignoring GOOGLE_CLOUD_LOCATION in .env as "--region" was explicitly passed and takes precedence
Initializing Vertex AI...
Vertex AI initialized.
Created sample_agent_tmp20260505_074456/agent_engine_app.py
Files and dependencies resolved
Deploying to agent engine...
INFO:vertexai_genai.agentengines:Creating in-memory tarfile of source_packages
INFO:vertexai_genai.agentengines:Using agent framework: google-adk
WARNING:google.auth.compute_engine._metadata:Compute Engine Metadata server unavailable on attempt 1 of 3. Reason: timed out
WARNING:google.auth.compute_engine._metadata:Compute Engine Metadata server unavailable on attempt 2 of 3. Reason: timed out

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecretsClient
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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Day 5b - Agent Deployment

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+ - Run All Code Sesión de apagado (ejecuta una celda 0 m borrador para comenzar)

```
# Initialize Vertex AI
vertexai.init(project=PROJECT_ID, location=deployed_region)

# Get the most recently deployed agent
agents_list = list(agent_engines.list())
if agents_list:
    remote_agent = agents_list[0] # Get the first (most recent) agent
    client = agent_engines
    print(f"✅ Connected to deployed agent: {remote_agent.resource_name}")
else:
    print("❌ No agents found. Please deploy first.")
```

WARNING:google.auth.compute_engine._metadata:Compute Engine Metadata server unavailable on attempt 1 of 3. Reason: timed out
WARNING:google.auth.compute_engine._metadata:Compute Engine Metadata server unavailable on attempt 2 of 3. Reason: timed out
WARNING:google.auth.compute_engine._metadata:Compute Engine Metadata server unavailable on attempt 3 of 3. Reason: timed out
WARNING:google.auth.compute_engine._metadata:Compute Engine Metadata server unavailable on attempt 1 of 5. Reason: [Errno -2] Name or service not known
WARNING:google.auth.compute_engine._metadata:Compute Engine Metadata server unavailable on attempt 2 of 5. Reason: [Errno -2] Name or service not known

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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Day 5b - Agent Deployment

Borrador guardado

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+ - Ejecutar todo Code Sesión de Cargando 10 m borrador

```
async for item in remote_agent.async_stream_query(
    message="What is the weather in Tokyo?",
    user_id="user_42",
):
    print(item)
```

Traceback (most recent call last)
/tmp/ipykernel_49/1869580144.py in <cell line: 1>()
----> 1 async for item in remote_agent.async_stream_query(
 2 message="What is the weather in Tokyo?",
 3 user_id="user_42",
 4):
 5 print(item)
NameError: name 'remote_agent' is not defined

+ Code + Markdown

What happened:

Misterios

Fragmento de código

```
from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret_value_0
```

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📄

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▶▶ Ejecutar todo

Code

Sesión de borrador

Cargando 10 m

...

incurring additional costs:

▶

agent_engines.delete(resource_name=remote_agent.resource_name, force=True)

print("✅ Agent successfully deleted")

Traceback (most recent call last)

/tmp/ipykernel_49/251702097.py in <cell line: 0>()

----> 1 agent_engines.delete(resource_name=remote_agent.resource_name, force=True)

2

3 print("✅ Agent successfully deleted")

NameError: name 'remote_agent' is not defined

+ Code + Markdown

What happened:

This cell deletes your deployed agent:

• resource_name: agent resource name. Identifier which agent to delete.

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr

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▶▶ Ejecutar todo

Code

Sesión de borrador

Cargando 10 m

...

The deletion process typically takes 1-2 minutes. You can verify deletion in the [Agent Engine Console](#).

✅ Congratulations! You're Ready for Production Deployment

You've successfully learned how to deploy ADK agents to Vertex AI Agent Engine - taking your agents from development to production!

You now know how to deploy agents with enterprise-grade infrastructure, manage costs, and test production deployments.

📖 Learn More

Refer to the following documentation to learn more:

- ADK Documentation
- Agent Engine Documentation
- ADK Deployment Guide

Other Deployment Options

Misterios

Fragmento de código

from kaggle_secrets import UserSecrets
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secr

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Competiciones

Puntos de referencia

Arena de juegos

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Más

Buscar

Iniciar sesión

Registro

Día 5 (Del prototipo a la producción)

Bienvenidos al Día 5.

Este documento técnico proporciona una guía técnica sobre el ciclo de vida operativo de los agentes de IA, centrándose en la implementación, el escalado y la puesta en producción. Explora los desafíos de la transición de los sistemas de agentes desde prototipos a soluciones de nivel empresarial, con especial atención al Protocolo Agente a Agente (A2A).

En los laboratorios de código, aprenderá a crear sistemas de múltiples agentes independientes que pueden comunicarse y colaborar mediante el Protocolo A2A. También aprenderá a llevar su agente desde su máquina local a un servicio escalable y listo para producción, implementándolo en Vertex AI Agent Engine en Google Cloud.

Tarea final:

Complete la Unidad 5 - "Del prototipo a la producción".

- Escucha el episodio del podcast que resume esta unidad.
- Para complementar el podcast, lea el informe técnico "Del prototipo a la producción".
- Completa estos codelabs en Kaggle:
 - Descubre cómo usar el protocolo A2A para que los agentes interactúen entre sí.
 - [Opcional] Implemente su agente en Agent Engine en Google Cloud.

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